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REVIEW

Effects of Resisted-Sprint Training on Sprint Performance and Mechanics: A Systematic Review and Meta-Analysis Focusing on Load Magnitude

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ABSTRACT

Resisted sprint training (RST) is effective for improving sprint acceleration. However, the choice of RST loads—ranging from light (≤10% of unresisted sprint maximum velocity decrement [v_{dec}], <20% of body mass [BM]) to very heavy (≥50% v_{dec} or ≥80% BM)—remains debated. To (i) explore the effects of different RST loads and unresisted sprint training (UST) on sprint performance and macroscopic mechanical outputs (e.g., theoretical maximal horizontal force [F_0], power [P_{max}], maximal ratio of horizontal-to-resultant force [RF_{max}]), and (ii) investigate how participant characteristics and training parameters influence sprint performance at varying RST loads. A search of 10 databases was conducted until September 2024, with studies included if they involved longitudinal interventions using horizontal resistance (e.g., sled) and if RST was the primary difference in training between groups. Analyses included single-arm (RST pre vs. post, considered exploratory only), two-arm (RST vs. UST), and multi-arm comparisons (different RST loads, UST, and control). Analytical models included two-level, three-level, and Bayesian-based network meta-analysis. The effects of different moderators on sprint performance and mechanical characteristics were explored through subgroup analyses and meta-regressions. Certainty of evidence was assessed using the GRADE approach. A total of 49 RST studies (1281 participants: 1153 males, 160 females, 64 unclear) were included. Pooled results showed RST was more effective than UST for improving sprint performance (effect size [ES] = −0.30, Hedges' g, Moderate GRADE). However, only moderate (≥20%–50% BM or > 10%–30% v_{dec}) and very heavy RST were more effective than UST, whereas light and heavy loads (> 50%–80% BM or > 30%–50% v_{dec}) were not. Two- and multiple-comparison models indicated that very heavy RST loads produced significantly greater performance improvements than light loads. The effect of RST on performance compared to UST became more pronounced with an increasing load ($\beta = -0.005$, $p = 0.018$), with a significant threshold at 20.7% BM, beyond which RST led to significantly greater performance improvements than UST ($p < 0.05$). Very heavy RST loads were particularly effective in improving F_0, P_{max}, and RF_{max} compared to UST, with adaptations in F_0 increasing significantly in tandem with RST ($\beta = 0.010$, $p = 0.023$). These effects were more pronounced in trained and highly trained individuals, with negligible improvements in recreationally active participants. There was an inverted U-shaped relationship between training volume

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(total distance) and training effect: heavy loads at small volumes yielded similar benefits to light loads at large volumes. Finally, participant characteristics (age, sex) and training parameters (duration, frequency, number of training sessions, sprint phase, session volume) also influenced sprint performance improvements. Very heavy loads were generally more effective at increasing sprint performance than UST and light RST loads, with light RST showing comparable effects to UST. These findings were likely attributable to improvements in the early acceleration phase (especially 0–5 m). Heavier loads were optimal for trained and highly trained groups, but no clear benefit of higher loads for recreational groups, suggesting light RST or UST are probably sufficient. Coaches and practitioners can refine protocols based on these findings to boost sprint performance outcomes.

Trial Registration: The original protocol was prospectively registered (osf.io/fu82d) with the Open Science Framework

1 | Introduction

Sprint performance is crucial for success in many individual (e.g., track and field) and team sports (e.g., soccer, rugby). Faster sprints offer advantages like improved speed and movement efficiency in individual sports [1, 2] and an ability to outpace opponents in team sports [3, 4]. Linear sprints are commonly divided into three phases: acceleration, maximum velocity, and deceleration [5–9]. However, in many team sports (e.g., baseball, soccer, and rugby), the maximum velocity phase is rarely reached (i.e., typically achieved 20–50 m from a stationary start), with repeated short sprints (i.e., < 20 m) being more common [4, 7, 10]. Therefore, rapid acceleration is often more important for sprint performance in these activities [3, 7, 9, 11]. Considering the distinct biomechanical characteristics of the acceleration and maximum velocity phases [5, 6, 12, 13], coaches frequently employ specific training methods to enhance acceleration.

From a kinetic perspective, the acceleration phase relies on the recruitment of lower limb muscle groups to produce large forces at foot contact with the ground [5–7, 14, 15]. Accordingly, training methods such as high-load resistance training and plyometrics can increase the force produced by the prime movers during acceleration [16–18]. However, most exercises involve bilateral, acyclical movements that lack specificity to running mechanics, which may limit their transferability [19, 20]. Since sprinting involves alternating unilateral, cyclic leg movements, the horizontal displacement of the center of mass is largely determined by the coordinated force exertion of the entire body, governed by the nervous system [9, 20–22]. Applying an external overload specific to sprint motion may more effectively transfer to sprint performance [7, 8, 23]. In addition to providing athletes with more track/field time, this approach blends the development of technical and physical capacities. This principle of "dynamic similarity" renders resisted-sprint training (RST) an attractive method to target improvements in sprint performance, particularly during the acceleration phase [8, 9, 18, 19, 24].

Several cross-sectional RST studies have shown that RST elicits greater hip and knee extensor muscle activation [25, 26], a more forward-leaning trunk [27, 28], and higher horizontal ground reaction forces compared to unresisted sprint training (UST) [29, 30]. Long-term (i.e., 4–11 weeks) RST significantly improves sprint performance (reduction in time to cover certain distances) during acceleration [8, 9]. Furthermore, RST enhances mechanical characteristics during acceleration, including theoretical maximal horizontal force [F_0], velocity [v_0], power output [P_{max}], the ratio of horizontal-to-resultant force [RF_{max}], the slope of the linear force-velocity relationship [S_{Fv}], and the rate of decrease in the ratio of horizontal-to-resultant force [DRF] [31–34]. For example, Stavridis et al. [35] demonstrated significant increases in F_0, P_{max}, RF_{max}, and 0–30 m sprint performance in sprinters after 6 weeks of twice-weekly, very heavy RST (50% of maximal sprint velocity decrement [$\%v_{dec}$]), although no statistically significant effects were observed on v_0 or DRF. Notably, light-loaded RST (e.g., 10% of body mass [BM]) did not significantly impact sprint performance or mechanical characteristics. These findings highlight that the magnitude of the RST load directly influences sprint performance and mechanical characteristics during the acceleration phase. Notably, RST load is commonly reported using %vdec, %BM, or absolute weight (kg) [32, 35–38]. Although these units reflect different aspects of resistance, they often correlate within a specific range (e.g., 10% vdec generally corresponds to 10%–13% BM) [8, 9, 39]. Due to inconsistencies in early RST studies [8, 18, 40, 41], defining a uniform standard is challenging. For clarity, we refer to all these measures collectively as "load" in this review.

From a kinematic perspective, some studies recommend limiting RST loads to low resistances (e.g., < 10% of maximum velocity reduction, or ~12.5% BM) [19, 23, 27, 40]. The primary rationale is the acute "disruption" of segment kinematics during resisted sprints and the hypothesized adverse effects of heavier loads on sprint technique, potentially reducing long-term performance or increasing injury risk [10, 19, 27]. Some research has therefore concluded that increasing RST load substantially influences kinematic variables [42–44], at least when compared acutely with to "control" conditions (e.g., free sprinting). Such evidence is often cited as the primary argument against heavy RST loads (i.e., ≥ 80% BM and ≥ 50% v_{dec}) [41, 43–45], despite their widespread use in high-performance sport. Notably, Zabaloy et al. [45] leveraged the principle of training specificity to conclude that very heavy RST could result in a "Groucho" running pattern where the knees remain flexed throughout the stride [45]. The authors suggested classifying very heavy RST as a tertiary training method, akin to gym-based strength and plyometric training, due to its limited specificity. However, a crucial point is that the substantial deviations from unresisted sprinting characteristics observed with increasing resistance (used as the basis for these arguments) are regularly compared across sprint distances and thus fail to acknowledge that the athlete may be traveling at very different velocities. This is important, as sprinting determinants are known to vary across phases, which might require different training approaches.

Varying RST load may aid in targeting characteristics underlying specific sprint phases [46, 47]. As such, a more nuanced comparison of the acute effects of resistance during RST would account for this. Notably, a recent cross-sectional study [48] comparing spatiotemporal variables and direct ground

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reaction forces during maximal sprints with different RST loads (low to high resistances, with the heaviest > 30% v_{dec}) found that loading elicited only minor differences in sprint mechanics when accounting for velocity. Similarly, another recent publication [49], adopting a "phase-based" comparison and cluster analysis of similar spatiotemporal and direct force-plate measurements, came to similar conclusions: When adopting a balanced approach, sprinting mechanics during RST appear similar to unresisted sprinting when compared at the same velocity or phase (e.g., high resistance to early sprint phases). Accordingly, limiting RST loads to prevent acute alterations in sprint technique (i.e., during resisted sprints) may cause athletes and coaches to miss out on the potential benefits of heavy loads [14, 50–52].

Most concerns around the deleterious effects of RST load stem from acute cross-sectional studies. For technical measures, there are surprisingly only two long-term studies of sprinting technique [34, 53], with neither evidencing meaningful degradation [34, 53]. For example, Lahti et al. [34] found that 9 weeks of very heavy RST (50% and 60% v_{dec}) did not significantly alter unloaded sprinting 2D kinematics. Several studies [29, 30, 54] have found that heavy loading increases net horizontal force and propulsive impulse compared to light loading, but these improvements are primarily due to the orientation of force application and longer contact times rather than increased absolute force production [30]. Furthermore, longer contact times may disrupt early acceleration timing and coordination, potentially impairing sprint performance [41, 45]. This highlights the ongoing debate between lighter loads, likely preserving sprinting technique, and heavier RST loads, which enhance both lower limb force generation and horizontal ground reaction force orientation. Taken together, these findings suggest that while long-term kinematic alterations have often been emphasized, they appear relatively minor and not clearly detrimental, particularly over time. Therefore, this review focuses primarily on sprint performance outcomes and kinetic adaptations (e.g., F_0, P_{max}, RF_{max}) as the most direct indicators of force-oriented adaptation. Discussion of kinematic changes is included mainly to provide context on load magnitude rather than as a central outcome.

Recent evidence suggests that heavier loads in RST may be more effective than lighter loads for improving acceleration. For example, when corrected for data entry errors [55] Alcaraz et al.'s [41] meta-analysis shows greater standardized mean differences in sprint performance for heavier RST (−0.71) compared to lighter loads (−0.54). Likewise, a recent review [9] found a higher standardized mean difference for very heavy (−1.01) compared to light RST (−0.69) in early acceleration, though this difference was not statistically significant ($p = 0.36$). Conversely, Hamad et al. [39] found that very heavy RST significantly outperforms lighter loads in sprint performance, regardless of sprint distance ($p < 0.01$). Nonetheless, the limited number of studies examining high resistances remains a persistent limitation, with the issue compounded by some omitting key articles [56]. Therefore, further analysis of the effects of RST loads could refine how coaches and sport scientists design RST training programs.

Another important issue in previous meta-analyses of RST [8–10, 18, 39, 41, 57] is the dichotomization of moderators that are continuous in nature. For example, Alcaraz et al. [41] categorized RST with 20% BM as the threshold. This approach has several drawbacks (1) valuable information is lost, increasing the risk of false negative; (2) it may under- or overestimate differences between categories, making variables near the cut-off appear more distinct than they are; (3) it can obscure potential linear or nonlinear relationships between variables and outcomes; and (4) it allows researchers to potentially select cut-offs based on pooled results, which could lead to erroneous outcomes, even when the cut-offs are predetermined. Therefore, some researchers have argued that continuous variables should not be dichotomized [58–61]. Moreover, existing meta-analyses on RST have generally failed to account for the hierarchical structure and correlation among effect sizes [8–10, 18, 39, 41, 57], and have rarely examined the linear or nonlinear effects of continuous moderators. Such limitations may produce less precise pooled estimates and limit our understanding of how continuous moderators influence the overall effects of RST.

Considering the limitations of two recent systematic reviews [9, 39], one focusing exclusively on team athletes and the other having methodological issues, and the publication of several recent original studies [31, 35, 62, 63], a more comprehensive meta-analysis of RST is needed. This analysis should employ advanced analytical methods to better understand the relationship between RST load and continuous variables. Therefore, the aims of this review were: (i) to investigate the effects of different RST loads on sprint performance and macroscopic mechanical characteristics across different comparison modalities (e.g., RST pre vs. post, RST vs. UST, Figure 1), and (ii) to examine how participant characteristics (e.g., training level, age, sex), sprint distances (e.g., 0–10, 0–20, 5–10, 10–20 m), and RST training parameters (e.g., duration, frequency, session volume) influence sprint performance across different RST loads.

2 | Methods

2.1 | Registration of Systematic Review Protocol

The systematic review followed the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) 2020 guidelines [64], and specific methodological details are provided in Appendix S1 of the electronic Supporting Information: Material (ESM). Study titles were preregistered in the Open Science Framework (osf.io/fu82d) after study retrieval and before data analysis.

2.2 | Eligibility Criteria

Inclusion criteria were adapted from previous RST systematic reviews [8, 9, 39, 41]. (i) Participants: healthy individuals of any sex or age, with at least a recreationally active training level [65]; (ii) Intervention: a minimum of 4 weeks. For single-arm comparisons, horizontal RST (e.g., sled, 1080Sprint) must be one of the training methods. In the two-arm comparisons, the key difference between groups should be the use of RST. Multi-arm comparisons could include RST versus UST, variations in RST loads, and control groups; (iii) Data: studies must provide descriptive participant details (e.g., age, body mass, training level) and

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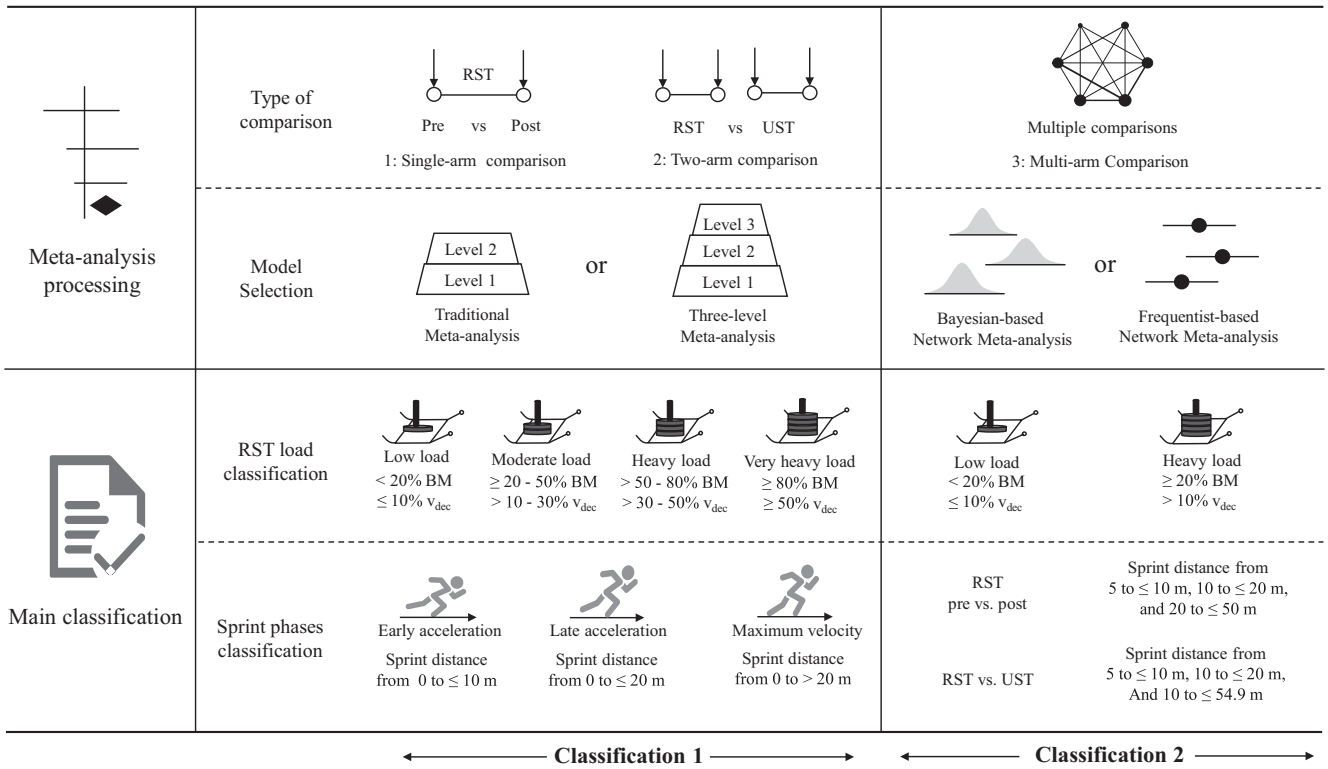

FIGURE 1 | Specific details of the article analysis process and major classifications (loads are first categorized according to $\%v_{dec}$, if $\%v_{dec}$ is not met, %BM prevails), RST, resisted sprint training; UST, unresisted sprint training; v_{dec}, velocity decrement; BM, body mass.

information on RST load and training volume; (iv) Comparison groups: single-arm RST requires no comparison group. For two-arm comparisons, the second group should be an unloaded UST group. Multi-arm comparisons may involve UST, different RST loads, and a control group with no specific resistance training; (v) Outcomes: sprint performance or sprint mechanics (e.g., F_0, v_0). RST should be performed on non-smooth surfaces (e.g., grass, synthetic track), excluding ice and treadmills; (vi) Testing: pre- and post-intervention sprint times must use automated electronic methods (e.g., timing gates, radar gun) or manual methods (e.g., stopwatch).

Following Cochrane Handbook's recommendations, no language restrictions were applied to maximize study inclusion. Conference proceedings and dissertations were included if they met inclusion criteria and provided complete information.

2.3 | Search Strategy

Search terms were adapted from previous RST systematic reviews [9, 39]. To ensure comprehensive coverage, searches were conducted in five English databases (Web of Science, SPORTDiscus, Embase, PubMed, and Scopus), one Chinese database (CNKI), and four gray literature databases (ClinicalTrials.org, MedRxiv, SportRxiv, ProQuest). Additional studies were identified by reviewing published RST reviews, utilizing Google Scholar, and examining citations of included studies. The search was independently conducted by two reviewers (K.X. and Y.M.Y.) with no language restrictions, covering the period from database inception to March 15, 2024. Boolean phrases and keywords used are detailed in Appendix S2 of the ESM. The final search was performed on September 15, 2024.

2.4 | Study Selection

One reviewer (K.X.) initially identified relevant studies and determined which titles and abstracts to include. During the eligibility phase, two reviewers (K.X. and Y.M.Y.) independently assessed the full texts based on the inclusion and exclusion criteria. Inter-rater agreement for full-text screening was high (Cohen's $\kappa = 0.88$), and disagreements were resolved through discussion until consensus. In cases of uncertainty, a third reviewer (R.W.) was consulted for the final decision. All processes were conducted using EndNote reference manager (version 20; Clarivate Analytics, Philadelphia, PA, USA).

2.5 | Data Extraction

Data extraction was independently conducted by two reviewers (K.X. and Y.M.Y.), with the extracted data reviewed and validated by two additional reviewers (W.J.T. and R.W.). Data from the included studies were organized in Excel (Microsoft Corporation, Redmond, WA, USA) under the following categories: (i) basic information: authors, publication date, sample size, sex, age, height, body mass, sports event, and season; (ii) intervention information: RST load, sprint distance per RST, session volume, weekly and total training volume, frequency, number of sessions, and durations; (iii) outcomes: sprint distance, sprint performance, and mechanical variables (F_0, v_0, P_{max}, RF_{max},

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DRF, and S_{FV}); (iv) other information: non-RST training, measurement devices, training devices, training surface, and RST attachment points.

If key data were insufficient or unclear (e.g., whether the comparison group was controlled or UST), the study authors were contacted via email or other means. For studies presented as images, data were extracted using WebPlotDigitizer (version 4.5; https://www.colliseum.net/WebPlot/) [66].

2.6 | Data Coding

Participant sex was categorized as female (if any female participants were included in the intervention group) or male (if all participants were male). Training levels were classified into three categories [39, 41]: (i) recreationally active, as individuals not engaged in regular sport practice or competition; (ii) trained, as individuals practicing and competing at an amateur club, university, or college, or otherwise not explicitly identified as elite or professional; (iii) highly trained, as those practicing and competing professionally or semi-professionally [65]. Sprint performance was divided into early acceleration, late acceleration, and maximum velocity phases, as defined in recent systematic reviews [8, 9, 41] (see Figure 1). RST load was categorized using two classification methods, as shown in Figure 1. Specifically, classification 1 divides RST loads into 4 categories (low: $<20\%$ BM/$\leq 10\%$ vdec; moderate: $\geq 20\%$–80% BM/$>10\%$–30% vdec; heavy: $>50\%$–80% BM/30%–50% vdec; very heavy load: $\geq 80\%$ BM/$\geq 50\%$ vdec) [39, 45], and classification 2 divides RST loads into 2 categories (low: $<20\%$ BM/$\leq 10\%$ vdec; heavy: $\geq 20\%$ BM/$>10\%$ vdec) [41]. For studies reporting $\%v_{dec}$ to %BM, a linear equation from Cahill et al. [67, 68] was used to convert between the two measures. Data coding was conducted in a blind manner by one author and two reviewers, two of whom were not directly involved in the project. Inter-rater reliability for data coding was excellent (intra-class correlation coefficient [ICC] = 0.92), and discrepancies were resolved through discussion until consensus. For studies reporting sprint performance as mean velocity, sprint time was calculated by dividing distance by mean velocity.

Additionally, regarding the calculation of RST training volume, the specific methods are as follows: the single-session training volume is determined by multiplying the distance of each RST sprint by the number of training sets; the weekly training volume is calculated by multiplying the number of weekly training sessions by the single-session training volume; the total training volume is the product of the single-session training volume and the total number of training sessions. Considering that some studies [69, 70] have adopted progressive training protocols, the training volume is categorized into minimum, average, and maximum values.

2.7 | Risk of Bias Evaluation and Certainty Assessment

Risk of bias was assessed using the Cochrane risk-of-bias tool (Rob-2) [71], based on the following criteria: (i) randomization process: a "low" risk was assigned only if both allocation concealment and specific randomization methods were reported; otherwise, the risk was considered "high"; (ii) deviations from intended interventions: since participants were aware of their training load during the RST intervention, the risk was considered as "low" if the study program was not influenced by other factors. If external factors affected the program, the risk was deemed "high"; (iii) missing outcome data: a "low" risk was assigned if there was no attrition or if attrition was addressed using an intention-to-treat approach [72]. Otherwise, the risk was classified as "high"; (iv) measurement of the outcome; and (v) selection of the reported result was based on the methodology of prior clinical registration: if clinical registration was available and reported results matched the registered results, the risk was considered "low"; otherwise, it was classified as "high". If there was no clinical registration, the risk was categorized as "some concerns."

Because participants in RST studies are typically aware of their intervention, we also assessed the risk of bias using the ROBINS-I tool for non-randomized studies. This tool evaluates seven domains: (i) bias due to confounding; (ii) bias in the selection of participants; (iii) bias in the classification of interventions; (iv) bias due to deviations from intended interventions; (v) bias due to missing data; (vi) bias in the measurement of outcomes; and (vii) bias in the selection of the reported result. Each domain was rated as "low," "moderate," "serious," or "critical" risk of bias. Detailed evaluation criteria are provided in Appendix S6.

The risk of bias in interpreting the results was assessed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) system [73]. Considering the results of the risk of bias assessments (Rob-2: 100% high risk; ROBINS-I: 44.90% high or serious risk), the initial evidence level for all outcomes was downgraded by one level to *moderate*. Subsequently, the evidence was downgraded to *low* or *very low* based on the following limitations: (i) insufficient statistical power was used as a proxy for sample size imprecision, (ii) one level was downgraded if the statistical power of the combined ES was $<80\%$, (iii) one level was downgraded if there was no clear direction in ES, (iv) two levels were downgraded if both issues were present, (v) high ($I^2 > 50\%$) statistical heterogeneity (inconsistency) resulted in a one-level downgrade, (vi) one level was downgraded if there was a risk of publication bias or if the number of combined ES was ≤ 2. Due to the uncertainty in determining an appropriate sample size, we were unable to specify what constituted a sufficient sample size. Hence, we employed statistical power as a substitute for sample size imprecision. This approach allowed us to factor in the potential impact of an inadequate sample size on the precision of our results [74, 75]. When the evidence is downgraded multiple times due to the aforementioned reasons, resulting in its quality falling below "very low," we consider its reliability insufficient to support any conclusions. Therefore, in this article, such evidence is classified as "no evidence" level (i.e., unranked) [76]. The same criteria were applied to rate the level of evidence for each group in all subgroup analyses. Statistical power for each pooled result was calculated using the *metameta* package [77].

The evaluation was conducted independently by two reviewers (K.X. and Y.M.Y.), with excellent inter-rater reliability for "RoB-2" (ICC = 0.98), "ROBINS-I" (ICC = 0.89), and GRADE

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assessments (ICC = 0.99). Disagreements were resolved through discussion until consensus. The Rob-2 results were visualized through the *robvis* package [78].

2.8 | Statistical Analysis

2.8.1 | Data Synthesis and Effect Measures

Three comparison types were included (Figure 1). For single-arm comparisons, the standardized mean change (SMC), adjusted for small samples, was used [79, 80]. To calculate the effect size for the single-arm study, a correlation between pre- and post-performance tests is required. However, as only one study reported a correlation coefficient r between 0.56 and 0.79 [62]. In this study, we assumed $r = 0.6$ for the calculations and conducted sensitivity analyses with r ranging from 0.5 to 0.9. For two-arm comparisons, it was considered that even small baseline differences can lead to a large change in performance. For example, a 4.1% baseline difference in 5-m sprint performance was observed (1.46 vs. 1.40) [35]. Therefore, comparisons were based on baseline-adjusted changes, and effect sizes were calculated as Hedges' g (small-sample corrected) [81].

When standard errors (SE) were reported, SD was calculated using: $\mathrm{SD} = \mathrm{SE} \times \sqrt{N}$. ES and SMC values were categorized as *small* (< 0.2), *moderate* (0.2–0.49), and *large* (0.5–0.8) [82]. Heterogeneity was evaluated using I^2, τ^2, and Q tests, with I^2 of 25%, 50%, and 75% indicating *low*, *moderate*, and *high* heterogeneity, respectively. The Q test was considered significant at $p < 0.10$. These indicators represent both relative and absolute values of residual heterogeneity, indicating the amount of unaccounted variability attributed to residual heterogeneity [83]. Prediction intervals (PI) were calculated to better account for potential variability of similar studies in the future [84]. A mixed-effects model with study as a random effect was used.

In the traditional meta-analysis, assuming studies are independent, ES with dependencies were combined using the formula provided in the Cochrane Handbook (See ESM Appendix S3). To control sample size inflation when groups were involved in multiple comparisons, the sample size was adjusted by dividing it by the number of comparisons. For example, if a control group with a sample size of 15 was involved in two comparisons, its sample size for each comparison was reduced to 7.5 [85].

2.8.2 | Three-Level Meta-Analysis

Considering that 83.3% of the studies involved multiple sprint phases, strong correlations were observed between sprint measurements (e.g., 0–10 m and segmented 0–5 m sprint performance) [5, 86]. A three-level meta-analysis was conducted to account for these correlations and hierarchy effects [87, 88]. This approach included a variance–covariance matrix-based clustering-robust variance estimation according to RST load classification 1 (Figure 1) along with small-sample adjustments to correct for within-study standard errors for correlations between effect sizes [89, 90]. The correlation coefficient ρ between different sprint distances was assumed to be 0.6, with sensitivity analyses conducted at 0.4 and 0.8. Multiple measurements within the same study were nested, so the variance of the observed ESs was decomposed into sampling variance, within-study variance (level 2), and between-study variance (level 3) to account for within-study (or within-group) correlations [91, 92]. The three-level meta-analyses accounted for the hierarchical nature of the data (e.g., ESs nested within studies), thereby improving statistical power and providing a more accurate representation of the effects of RST on sprint performance [88, 93].

The decision to present models was based on a comparison of goodness-of-fit indices (e.g., Akaike Information Criterion [AIC], Bayesian Information Criterion [BIC]). When the three-level model was higher than that of the traditional two-level model, the three-level model was used, and *vice versa*. Model parameters were estimated using the Restricted Maximum Likelihood method, with observations weighted by the inverse of the sampling variance (See ESM Appendix S2 for details). Three-level meta-analyses were conducted using the *metafor* package in *R* (version 4.3.0; R Core Team, Vienna, Austria) [94], with robust methods for correlated and hierarchical effects implemented using the *clubSandwich* package [90].

2.8.3 | Subgroup Analysis and Meta-Regression

Subgroup analyses were conducted for dichotomous variables including participants' sex, age, training level, RST load classifications (1 and 2), and sprint distance. RST load and sprint phase classifications for continuous variables are commonly used in this field [8, 9, 39, 41, 45]. Initial subgroup analyses used single moderator variables, followed by two-moderator analyses with RST load classification 1 fixed as one of the moderators [94].

Only one previous RST-related meta-analysis explored the effect of a continuous variable (sprint distance) on sprint performance using meta-regression [39]. However, that study incorrectly applied simple linear regression as meta-regression, potentially leading to inaccurate estimates of regression coefficients and p-values [95], as explained in Section 4.2. In this study, meta-regression was performed when at least 10 ESs were available for continuous variables, including participant age, training duration, frequency, number of total sessions, sprint distance per single RST, weekly total and total training volumes, sprint performance test distance after RST, RST load, F_0, P_{max}, and RF_{max}. Meta-regression was also conducted based on RST load classification 1 for session volume, total volume, and sprint performance test distance after RST. Various timepoint models (linear, cubic polynomial, restricted cubic spline, natural cubic spline, and thin plate spline models) were tested using model heuristics (e.g., AIC, BIC) and visual inspection to determine the best fit [96]. Large-scale bound-constrained optimization was applied when model convergence failed [97]. Additionally, we identified the threshold where the 95% confidence interval of the meta-regression model crossed zero. Specifically, this corresponds to the point where the lower bound of the 95% confidence interval first exceeds zero or the upper bound first falls below zero in the linear model. The model selection and analysis process is available on the Open Science Framework (osf.io/fu82d).

For linear meta-regression, the average value of interval data was chosen as the continuous conditioning variable. For

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example, Lahti et al. [34] used RST loadings ranging from 95% to 120% BM, so a mean value of 107.5% BM was applied in the meta-regression. Sensitivity analyses were performed using the minimum and maximum values within each data interval to assess result stability. Meta-regression results for sprint mechanics were visualized using the *ggplot2* package [98].

Heavy loads (> 50%–80% BM or > 30%–50% vdec) were excluded from subgroup analysis due to the limited number of available studies in single-arm ($n = 4$) and two-arm comparisons ($n = 1$). Because RST loads are often reported as ranges (e.g., 45%–65% BM), we employed both meta-regression and subgroup analyses to more comprehensively investigate the effects of RST load on sprint performance and mechanical characteristics.

2.8.4 | Network Meta-Analysis

Several included studies reported intercomparisons between 3 and 4 different RST loads [35, 67, 68, 99, 100], complicating direct pairwise comparisons [101]. To explore the relationships between different RST loads, UST, and CG, a Bayesian arm-based multilevel network meta-analysis was conducted. This approach calculates pairwise effect sizes from studies comparing two treatments (direct evidence) and generates indirect evidence through a generic comparator [101]. This analysis was conducted using *gemtc* [102] and *BUGSnet* [103] packages, running Markov chain Monte Carlo simulations with four chains for 105 000 iterations (with an aging period of 5000). Convergence was assessed using the Gelman-Rubin method. To test the robustness of Bayesian model results, a frequency network meta-analysis was also conducted using the *netmeta* package [104]. Multiple ESs with dependencies within a single treatment were nested within treatment arms [76], assuming a rho of 0.6 (sensitivity analysis for correlations of 0.4 and 0.8, see ESM Appendix S3). The approach provided an accurate estimate of the average effect of each RST load on sprint performance.

To test the consistency assumption of the network meta-analysis, the fit of the base case model was compared to that of the inconsistency model. The Surface Under the Cumulative Ranking curve (SUCRA; [105]) was used to summarize differences in sprint performance across RST loads. For different RST loads, SUCRA values expressed as a percentage were calculated to indicate the likelihood of different RST loads improving sprint performance relative to UST. P-source was also used for comparison [106].

2.8.5 | Risk of Bias and Sensitivity Analyses

Sunset plots were used for single-arm and two-arm comparisons to evaluate publication bias and the statistical power of each ES [77]. A significant risk of publication bias was indicated if $p < 0.05$ using the Egger test for three-level meta-analysis [107]. For network meta-analysis, the *netmeta* package was employed to detect publication bias [104].

Leverage, outliers, and influence were assessed for all meta-analysis models by calculating hat values, Cook's distances, and studentized residuals, respectively [108–110]. Potential outliers were flagged if their hat values and Cook's distances exceeded three times the mean, and if studentized residuals were > 3. After excluding these outliers, calculations were re-run. If the results remained unchanged, the original model was retained; otherwise, the model excluding the outliers was considered [111].

Additionally, a sensitivity analysis was performed excluding studies that used stopwatch measurements. These studies were excluded, and analyses were repeated. If the exclusion led to a meaningful change in the interpretation of the effect magnitude, the adjusted model was adopted; otherwise, the original model was retained.

3 | Results

3.1 | Search Results

The initial search yielded 12 484 studies, which were reduced to 6178 after removing duplicates. After screening abstracts and titles, 188 studies remained, with 39 included after a full-text review. An additional 10 studies were found via supplementary searches, resulting in a total of 49 included studies (Figure 2). Five studies [21, 63, 100, 112, 113] presented data extracted using visual tools. We contacted the authors of two studies [114, 115] to clarify the details of the control groups. One author confirmed that the control group was a non-intervention group [114]; the other did not respond, so Harrison and Bourke's [115] control group was classified as a non-intervention group based on previous reviews and the article description. In addition, we contacted the author of another study to request data [116], but no response was received; relevant data were extracted from a published meta-analysis. Overall, these procedures did not affect the final analysis.

3.2 | Study Characteristics

Of the 49 included studies, 35 were randomized controlled trials, two were quasi-randomized, six were non-randomized, and five were before-and-after comparisons only (see ESM Appendix S4). The total sample size was 1281 (male: $n = 1153$, female: $n = 160$, unclear: $n = 64$), with ages ranging from 10.4 ± 0.8 to 27.1 ± 4.1 years. Participants included team athletes ($n = 39$), sprinters ($n = 2$), multiple-sport athletes ($n = 1$), and an unclear category ($n = 7$).

Studies were executed in-season ($n = 17$), pre-season ($n = 11$), off-season ($n = 6$), both pre-season and off-season ($n = 1$), both pre-season and in-season ($n = 1$), or otherwise the season was unclear ($n = 13$). Performance testing devices included timing gates ($n = 27$), radar guns ($n = 8$), motorized devices ($n = 1$), video analysis ($n = 4$), velocimeters ($n = 1$), velocimeters with a stopwatch ($n = 2$), stopwatches ($n = 5$), and timing gates and radar guns combined ($n = 1$).

Forty-seven studies used a towed horizontal resistance device, while two used a sled-push device (i.e., arms used to push). Training devices included sleds ($n = 43$), robotic/motorized resistance systems ($n = 2$), elastic resistance platforms ($n = 1$), and other customized or combined systems. Tethering points were mainly at the waist ($n = 28$) or shoulders ($n = 6$), with 15 studies not reporting this information. Thirty-five studies (71.4%)

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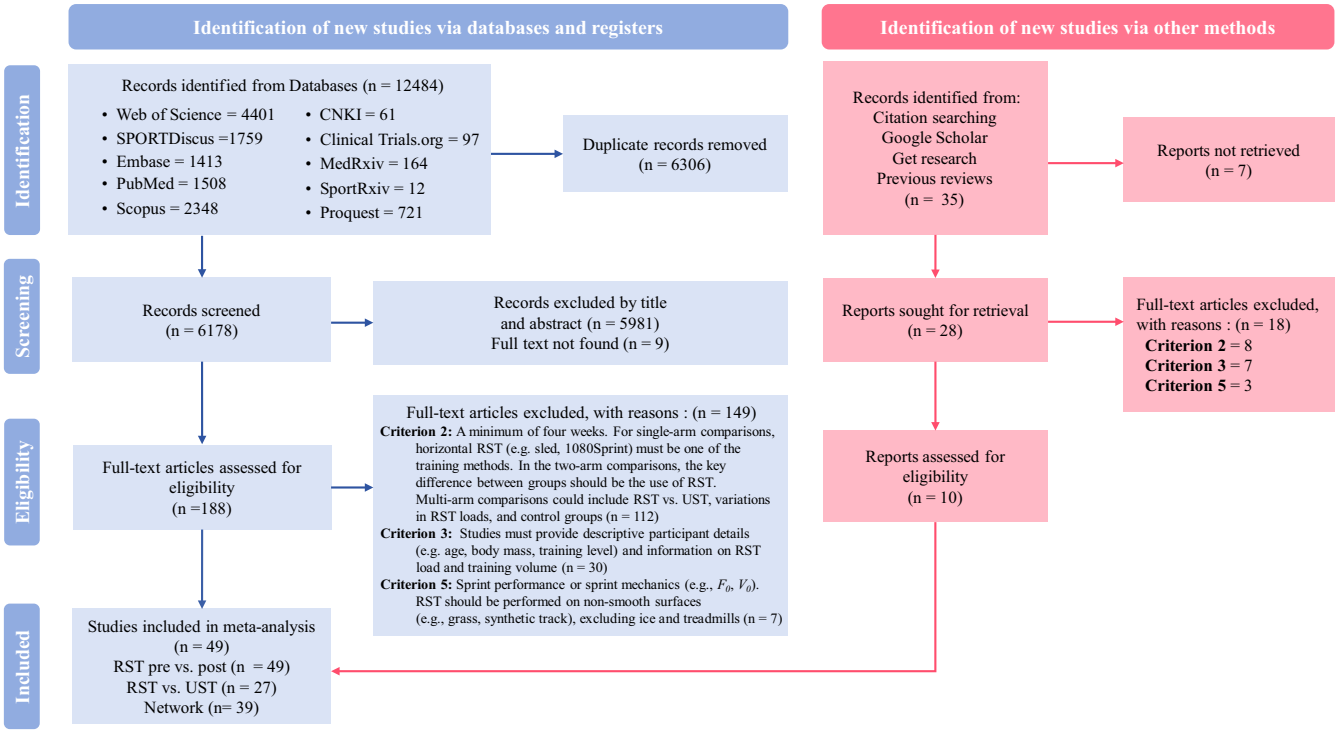

FIGURE 2 | PRISMA flow diagram detailing the study inclusion process.

incorporated other forms of training during the intervention, while 14 studies did not specify.

RST training parameters included single sprint distances (5–55 m), single-session training distances (30–600 m), total weekly training distances (60–1200 m), and total training distances (380–6720 m). The total number of training sessions ranged from 6 to 36, with a frequency of 1–3 sessions per week. Forty studies used or mentioned loads applied relative to BM (5–133[±13] %BM); 18 studies used or mentioned loads as a reduction in maximum velocity (5.6% v_{dec}–80% v_{dec}); and four studies used absolute loads (2.7–35 kg). Additionally, 13 studies reported sprint mechanics parameters including F_0 ($n = 13$), v_0 ($n = 10$), P_{max} ($n = 10$), RF_{max} ($n = 8$), S_{FV} ($n = 7$), and DRF ($n = 6$).

3.3 | Risk of Bias Assessment

Only two studies [62, 117] performed allocation concealment and reported specific randomization methods, classifying them as "low-risk" according to Rob-2's evaluation criteria. All other studies lacked allocation concealment and thus were classified as "high risk." No interventions were influenced by other factors during the intervention period, so deviations from intended interventions were deemed "low risk." Eighteen studies were classified as "high risk" due to sample attrition and failure to use an intention-to-treat approach, while the remaining studies were considered "low risk." No studies were pre-registered, leading to "some concern" regarding outcome measurement and result selection. Overall, all studies (100%) were ultimately classified as "high risk" of bias (see ESM Appendix S6, Figure S1), following the strict Rob-2 evaluation criteria.

For the ROBINS-I assessment, a total of 22 studies (44.90%) were rated as a serious risk of bias, and 27 studies (55.10%) at moderate risk. Detailed results and domain-specific ratings are provided in ESM (Appendix S6).

3.4 | Risk of Publication Bias and Sensitivity Analysis

A risk of publication bias was detected in single-arm comparisons ($p < 0.01$, ESM Appendix S7, Figure S2), likely because these studies, like cohort studies, often show improvement in sprint performance with prolonged training. Nonetheless, these findings in single-arm comparisons do not necessarily indicate true publication bias.

Funnel plots for two-arm comparison appeared more symmetrical, and Egger's test showed no significant publication bias ($p = 0.99$, ESM Figure S2). Likewise, Egger's test for multi-arm comparison showed no significant risk of publication bias ($p = 0.88$, ESM Figure S7).

Varying the correlation r in a two-arm study did not change the main results of the pooled model (see ESM Appendix S5). For detailed information on outlier exclusion and pooled model selection, see R code (osf.io/fu82d) and ESM (Appendix S4).

3.5 | Main Effect and Certainty of Evidence

3.5.1 | Single-Arm Comparison

RST pre- and post-intervention comparisons showed a moderate improvement in sprint performance (SMC = −0.50, 95%

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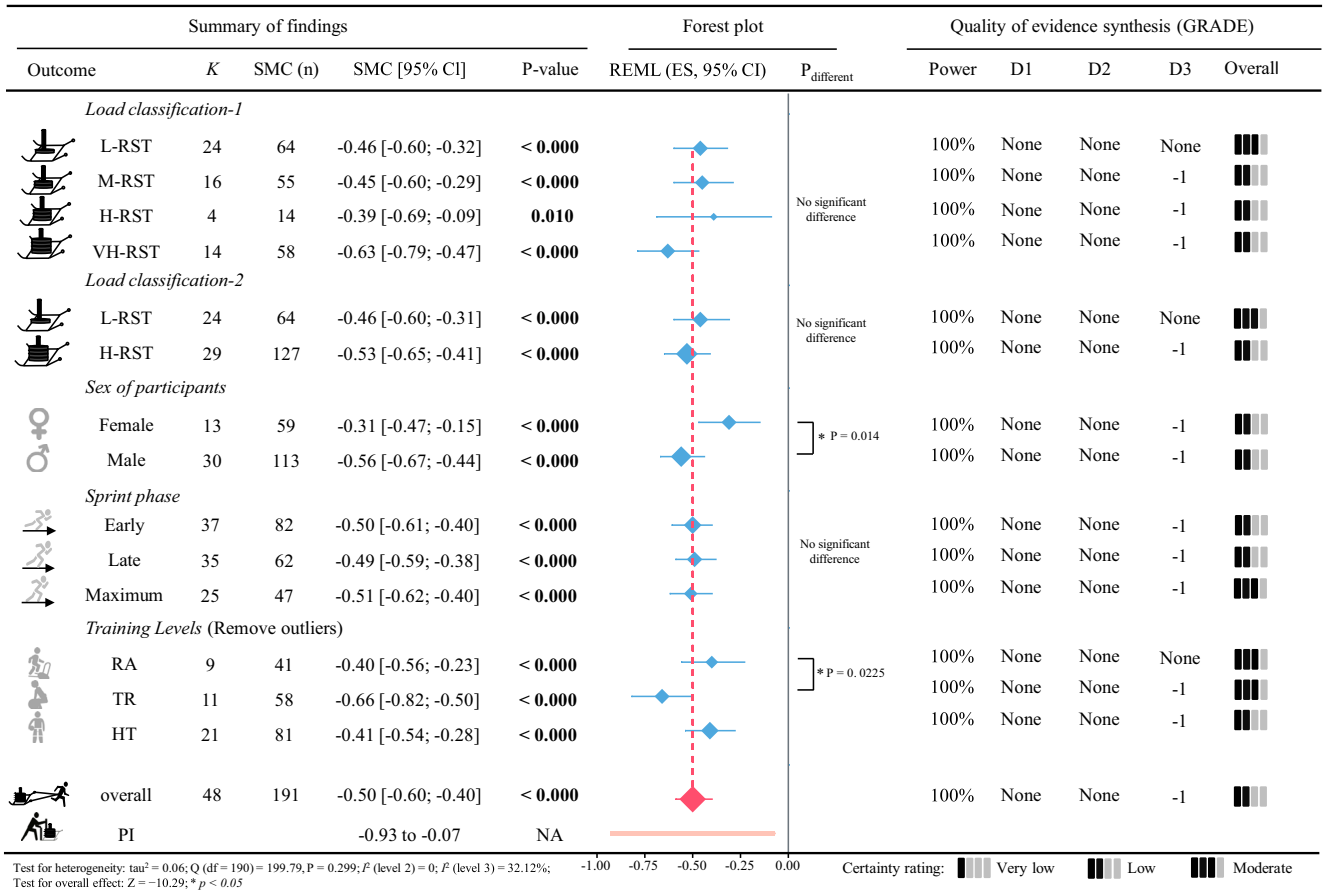

FIGURE 3 | Effect of RST pre versus post on sprint performance subgroup analysis results, D1, whether the statistical power reaches 80% and the effect size has a clear direction; D2, whether $I^2 > 50\%$; D3, whether there is a risk of publication bias; L, light; M, moderate; H, heavy; VH, very heavy; RST, resisted sprint training; RA, recreationally active; TR, trained; HT, highly trained; PI, prediction intervals; REML, restricted maximum likelihood; SMC, standardized mean change; CI, confidence interval.

CI [−0.60 to −0.40], $p < 0.001$, Q (190) = 199.79, $p = 0.299$, I^2-level 2 = 0%, I^2-level 3 = 32.12%, $\tau^2 = 0.045$, PI = −0.93 to −0.07, Statistical power = 100%, Low GRADE, Figure 3). The evidence for all subgroups ranged from very low to moderate.

RST significantly enhanced all sprint mechanics parameters (Figure 4). Subgroup analyses revealed the greatest improvements with very heavy RST on F_0 (SMC = 0.69), P_{max} (SMC = 0.55), RF_{max} (SMC = 0.74), DRF (SMC = −0.28), and S_{FV} (SMC = −0.33). However, no significant differences were observed for any mechanical characteristics ($p > 0.05$).

Subgroup analyses based on RST load classification 1 (Figure 1) showed that very heavy RST (SMC = −0.63) had a greater improvement in sprint performance than heavy (SMC = −0.39), moderate (SMC = −0.45), and light loads (SMC = −0.46). According to the RST load classification 2 (Figure 1), heavy RST (SMC = −0.53) had a greater improvement than light RST (SMC = −0.46) in sprint time. However, no significant differences were found between load groups in either classification method. The improvement of RST was greater ($p = 0.023$) in males (SMC = −0.56) than in females or mixed-sex groups (ES = −0.31). Additionally, RST had the strongest improvement in the trained group (SMC = −0.66) compared to the recreationally active (SMC = −0.40) and highly trained groups (SMC = −0.41) in sprint time.

RST had comparable improvements across sprint phases, including early acceleration (SMC = −0.50), late acceleration (SMC = −0.49), and maximal velocity phases (SMC = −0.51, Figure 3). Significant improvements in sprinting times ($p < 0.001$, Figure 5) were observed across various distances: 5–10 m (SMC = −0.33), 10-20 m (SMC = −0.31), and 20–50 m (SMC = −0.22). In the 5–10 m sprint phase, moderate RST (SMC = −0.41) was more effective than very heavy (SMC = −0.33) and light RST (SMC = −0.24). In the 10–20 m sprint phase, both light RST (SMC = −0.21, $p = 0.031$) and very heavy RST (SMC = −0.31, $p < 0.001$) led to statistically significant improvements in sprint time. Likewise, in the 20–50 m phase, light (SMC = −0.18, $p = 0.046$) and very heavy RST (SMC = −0.32, $p = 0.019$) showed significant effects in sprint time.

Further subgroup analysis based on RST load classification 1 (Figure 1) showed similarities to those of the single moderator subgroup analyses (e.g., participant sex and sprint phase) (Table 1). While light loads were more effective for recreationally active groups (SMC = −0.60), moderate (SMC = −0.72) and very heavy loads (SMC = −0.73) had a greater impact on trained groups (Table 1).

After excluding studies that relied on stopwatch measurements, the results remained essentially unchanged (SMC = −0.46; 95%

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Summary of findings						Forest plot		Quality of evidence synthesis (GRADE)				
Outcome		*K*	Study (N)	SMC [95% CI]	P-value	REML (ES, 95% CI)	$P_{different}$	Power	D1	D2	D3	Overall
V_0	L-RST	3	40	0.26 [-0.10; 0.62]	0.155		No significant difference	9.7%	-2	None	None	
	M-RST	2	29	0.22 [-0.20; 0.64]	0.313			6.9%	-2	None	-1	
	VH-RST	14	180	0.11 [-0.06; 0.28]	0.211			10%	-2	None	None	
	Overall	19	249	0.15 [0.01; 0.28]	**0.040**			18%	-2	None	None	
	PI			-0.11 to 0.41	NA							
F_0	L-RST	4	45	0.42 [0.11; 0.73]	**0.008**		No significant difference	23%	-1	None	None	
	M-RST	2	29	0.22 [-0.15; 0.59]	0.246			7.0%	-2	None	-1	
	VH-RST	15	188	0.69 [0.53; 0.85]	**< 0.000**			99%	None	None	-1	
	Overall	21	262	0.59 [0.44; 0.73]	**< 0.000**			99%	None	None	-1	
	PI			0.32 to 0.86	NA							
Pmax	L-RST	3	40	0.45 [0.12; 0.78]	**0.007**		No significant difference	24%	-2	None	None	
	M-RST	2	29	0.30 [-0.07; 0.67]	0.115			9.8%	-2	None	-1	
	VH-RST	15	188	0.62 [0.46; 0.78]	**< 0.000**			98%	None	None	-1	
	Overall	20	257	0.55 [0.42; 0.68]	**< 0.000**			99%	None	None	-1	
	PI			0.42 to 0.68	NA							
RF_{max}	L-RST	3	40	0.66 [0.31; 1.01]	**0.005**		No significant difference	44%	-1	None	None	
	VH-RST	10	132	0.76 [0.57; 0.96]	**< 0.000**			98%	None	None	None	
	Overall	13	172	0.74 [0.57; 0.91]	**< 0.000**			99%	None	None	None	
	PI			0.57 to 0.91	NA							
D_{RF}	L-RST	3	40	-0.16 [-0.48; 0.16]	0.335		No significant difference	6.0%	-2	None	None	
	VH-RST	6	84	-0.28[-0.51; -0.05]	**0.015**			23%	-1	None	None	
	Overall	9	124	-0.24 [-0.42; -0.06]	**0.010**			22%	-1	None	None	
	PI			-0.42 to -0.06								
S_{FV}	L-RST	2	27	-0.16 [-0.54; 0.22]	0.414		No significant difference	5.3%	-2	None	-1	
	VH-RST	8	98	-0.38[-0.59; -0.17]	**< 0.000**			41%	-1	None	None	
	Overall	10	125	-0.33 [-0.51; -0.15]	**< 0.000**			40%	-1	None	None	
	PI			-0.51 to -0.15	NA							

Test for heterogeneity: V_0 I^2 = 11.59%; F_0 I^2 = 10.39%; P_{max} I^2 = 0%; RF_{max} I^2 = 0%; D_{RF} I^2 = 0%; S_{FV} I^2 = 0%

-1.0 -0.5 0.0 0.5 1.0

Certainty rating: Very low Low Moderate

FIGURE 4 | Effect of RST pre versus post on sprint mechanics subgroup analysis results, D1, whether the statistical power reaches 80% and the effect size has a clear direction; D2, whether $I^2 > 50\%$; D3, whether there is a risk of publication bias; L, light; M, moderate; H, heavy; VH, very heavy; RST, resisted sprint training; RA, recreationally active; TR, trained; HT, highly trained; PI, prediction intervals; REML, restricted maximum likelihood; SMC, standardized mean change; CI, confidence interval; K, number of studies; Study (N), sample size; v_0, maximal theoretical running velocity; P_{max}, maximal power; F_0, maximal theoretic al horizontal force; RF_{max}, maximal ratio of force; DRF, decrease in the ratio of force; S_{FV}, slope of the linear force-velocity relationship.

CI [−0.56 to −0.36]; $p < 0.001$). Therefore, the original model including all studies was retained.

3.5.2 | Two-Arm Comparison

Compared to UST, RST showed a moderate improvement in sprint performance (ES = −0.30, 95% CI [−0.48 to −0.12], $p = 0.002$, Q (108) = 74.92, $p = 0.99$, I^2-level 2 = 0%, I^2-level 3 = 29.68%, $\tau^2 = 0.08$, PI = −0.89 to 0.28, Statistical power = 100%, Moderate GRADE, Figure 6). The overall level of evidence for all subgroups ranged from unranked to moderate.

RST significantly increased sprinting mechanics characteristics ($p < 0.05$, Figure 7), with notable improvements in F_0 (ES = 0.42), P_{max} (ES = 0.35), and RF_{max} (ES = 0.46). However, no significant improvements were noted for v_0 (ES = −0.04), DRF (ES = −0.31), and S_{FV} (ES = −0.29). Among the different load conditions, very heavy RST had the greatest improvements on F_0 (ES = 0.64), P_{max} (ES = 0.51), and RF_{max} (ES = 0.46), with RF_{max} showing a significant difference ($p = 0.037$) compared to light RST.

Subgroup analysis regarding RST load classification 1 (Figure 1) showed that very heavy RST (ES = −0.68) was more effective than both light (ES = −0.05; $p < 0.001$) and moderate loads (ES = −0.29; $p = 0.026$). The results of load classification 2 (Figure 1) showed that heavy RST (ES = −0.47) was more effective ($p < 0.05$) than light (ES = −0.04). Additionally, RST was more effective for males (ES = −0.37) than for females or mixed-sex groups (ES = −0.14). The improvement of RST was also greater for trained individuals (ES = −0.37) than for recreationally active (ES = −0.17) and highly trained groups (ES = −0.29).

Like the single-arm comparison, RST effects were consistent across different sprint phases. However, compared to UST, RST did not significantly increase sprinting times in the 5–10 m (ES = −0.20), 10–20 m (ES = −0.02), and 10–54.9 m (ES = 0.05) sprint phases. Additionally, no significant difference ($p > 0.05$) was noted between different RST loads.

Further subgroup analyses based on RST load classification 1 (Figure 1) showed that only very heavy loads had a significant

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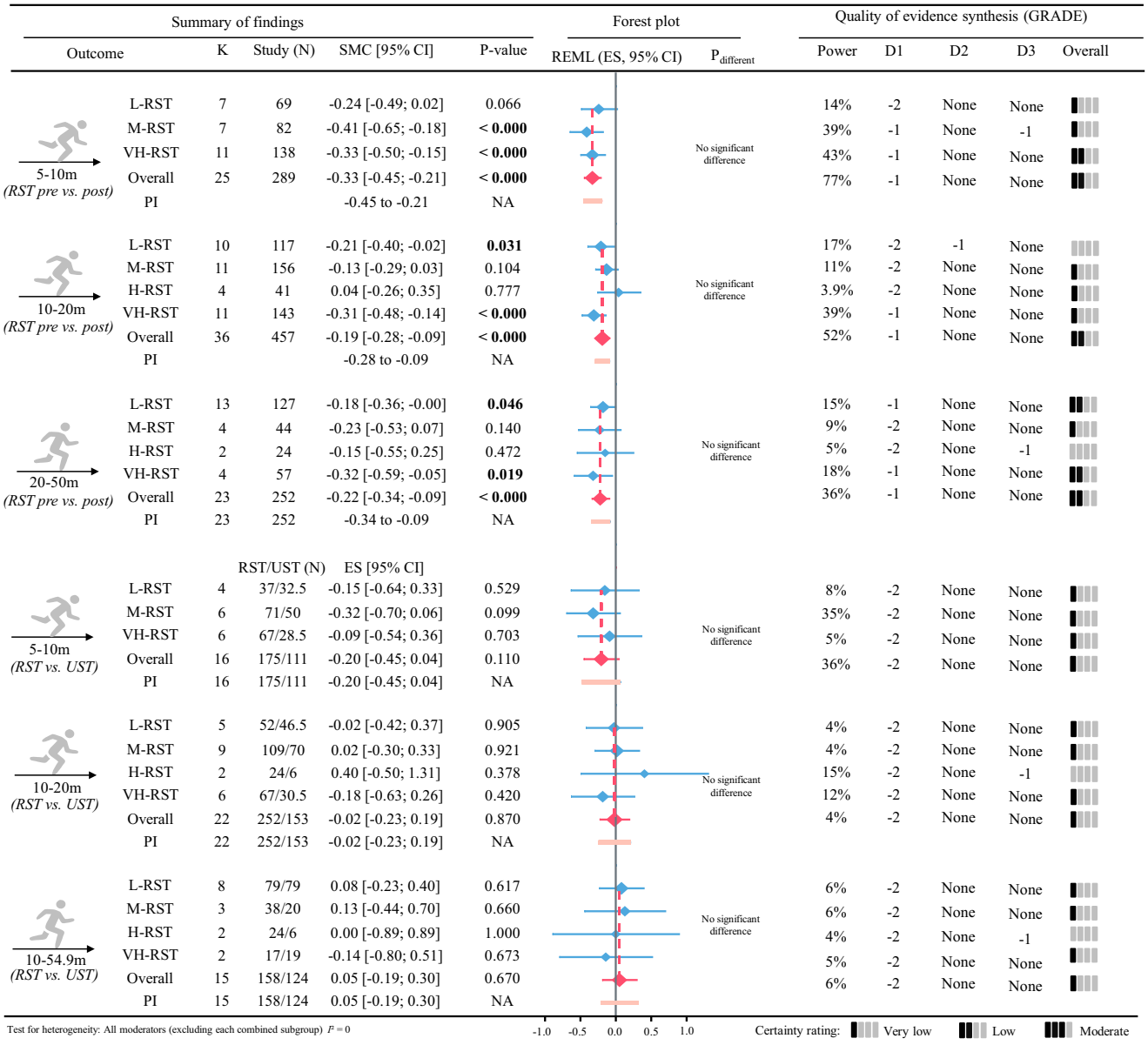

FIGURE 5 | Subgroup analysis of single-arm and two-arm comparisons on > 5 m segmental distance sprint performance, D1, whether the statistical power reaches 80% and the effect size has a clear direction; D2, whether $I^2 > 50\%$; D3, Whether there is a risk of publication bias; L, light; M, moderate; H, heavy; VH, very heavy; RST, resisted sprint training; UST, unresisted sprint training; RA, recreationally active; TR, trained; HT, highly trained; PI, prediction intervals; REML, restricted maximum likelihood; SMC, standardized mean change; CI, confidence interval; K, number of studies; Study (N), sample size; RST/UST (N), Sample size for RST and UST.

improvement on male groups (ES = −0.74, $p < 0.001$, Table 1). Regarding sprint phases, light loads did not outperform UST in any phase ($p > 0.05$), moderate loads were more effective than UST in the early sprint phase (ES = −0.33, $p = 0.011$), while very heavy loads were more effective than UST in all phases ($p < 0.05$). Regarding training level, light loads had the greatest improvements on recreationally active individuals, but this was not significant (ES = −0.37, $p = 0.120$). Moderate loads were most effective for trained participants (ES = −0.44, $p = 0.040$), while very heavy loads had the largest improvements on highly trained individuals (ES = −1.11, $p < 0.001$, Table 1).

After excluding studies that relied on stopwatch measurements, the results remained essentially unchanged (ES = −0.26; 95% CI [−0.44 to −0.09]; $p = 0.003$). Therefore, the original model including all studies was retained.

3.5.3 | Multi-Arm Comparison

Figure 8 illustrates a network diagram of pairwise effect sizes calculated for the four RST loads (light, moderate, heavy, and very heavy) compared to UST and control groups. A total of 39 studies with 84 groups and 973 participants were included. Only

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TABLE 1 | Subgroup analysis of moderators under different loading conditions.

Load	Moderator	RST pre vs. post			RST vs. UST		
		SMC (*n*)	SMC (95% CI)	*p*	ES (*n*)	ES (95% CI)	*p*
Participant sex							
Light	Female	23	−0.22 [−0.44 to −0.00]	**0.046**	15	0.11 [−0.33; 0.54]	0.630
	Male	30	−0.57 [−0.74 to −0.39]	**< 0.001**	14	−0.13 [−0.45; 0.19]	0.420
Moderate	Female	16	−0.38 [−0.67 to −0.10]	**0.009**	9	−0.26 [−0.75; 0.22]	0.280
	Male	37	−0.42 [−0.60 to −0.25]	**< 0.001**	26	−0.29 [−0.60; 0.02]	0.070
Very heavy	Female	17	−0.36 [−0.62 to −0.09]	**0.009**	10	−0.48 [−1.08; 0.13]	0.120
	Male	35	−0.74 [−0.94 to −0.53]	**0.001**	22	−0.74 [−1.06; −0.42]	**< 0.001**
Sprint phase							
Light	Early	23	−0.44 [−0.60; −0.27]	**< 0.001**	13	−0.07 [−0.34; 0.20]	0.610
	Late	18	−0.43 [−0.61; −0.26]	**< 0.001**	10	0.06 [−0.24; 0.35]	0.704
	Maximum	23	−0.50 [−0.66; −0.33]	**< 0.001**	11	−0.12 [−0.41; 0.16]	0.389
Moderate	Early	27	−0.44 [−0.61; −0.27]	**< 0.001**	20	−0.33 [−0.58; −0.08]	**0.011**
	Late	18	−0.45 [−0.62; −0.27]	**< 0.001**	14	−0.26 [−0.52; 0.00]	0.052
	Maximum	10	−0.50 [−0.70; −0.30]	**< 0.001**	3	−0.31 [−0.71; 0.08]	0.120
Very heavy	Early	27	−0.67 [−0.85; −0.50]	**< 0.001**	15	−0.73 [−1.02; −0.43]	**< 0.001**
	Late	20	−0.61 [−0.79; −0.43]	**< 0.001**	13	−0.68 [−0.97; −0.38]	**< 0.001**
	Maximum	11	−0.58 [−0.77; −0.38]	**< 0.001**	4	−0.56 [−1.00; −0.12]	**0.013**
Participant training level							
Light	RA	16	−0.60 [−0.84; −0.35]	**< 0.001**	5	−0.37 [−0.84; 0.10]	0.120
	TR	17	−0.46 [−0.71; −0.21]	**< 0.001**	13	0.07 [−0.31; 0.45]	0.720
	HT	31	−0.34 [−0.54; −0.14]	**< 0.001**	16	0.03 [−0.33; 0.40]	0.860
Moderate	RA	19	−0.36 [−0.58; −0.13]	**< 0.001**	9	0.02 [−0.41; 0.44]	0.940
	TR	10	−0.72 [−1.04; −0.39]	**< 0.001**	9	−0.44 [−0.87; −0.01]	**0.040**
	HT	26	−0.34 [−0.58; −0.09]	**0.007**	19	−0.38 [−0.71; −0.04]	**0.030**
Very heavy	RA	3	−0.21 [−0.74; 0.32]	0.430	0	NA	NA
	TR	33	−0.73 [−0.94; −0.52]	**< 0.001**	27	−0.62 [−0.90; −0.34]	**< 0.001**
	HT	22	−0.60 [−0.85; −0.35]	**< 0.001**	5	−1.11 [−1.80; −0.42]	**< 0.001**

Abbreviations: CI, confidence interval; ES, effect size; HT highly training; RA, recreationally active; RST, resisted sprint training; SMC, standardized mean change; TR, trained; UST, unresisted sprint training. *Note*: Bold values are statistically significant ($p < 0.05$).

very heavy RSTs were more effective than USTs (ES = −0.51, 95% credible intervals [CrI] −0.77 to −0.26, Table 2). When combining indirect and direct comparisons, very heavy RST was more effective than light (ES = −0.34, 95% CrI [−0.63 to −0.06]). There was some inconsistency between the very heavy RST and UST comparison ($p = 0.03$), albeit without compromising the stability of the Bayesian model. Furthermore, SUCRA and P-score results consistently showed that very heavy RST was the most effective for improving sprint performance, followed by moderate and light loads (Table 2).

The results of the frequency-based network meta-analysis aligned with those of the Bayesian-based analysis. Sensitivity analysis confirmed that the pooled results were unaffected by the correlation between RST sprint phases (see ESM Appendices S8 and S9, Figures S8–S18).

3.6 | Meta-Regression

3.6.1 | Single-Arm Comparison

Among the 18 continuous moderators of sprint performance, RST effects on sprint performance were significantly influenced by weeks ($\beta = -0.057$, $p = 0.048$, Figure 9B), training frequency ($\beta = -0.178$, $p = 0.050$, Figure 9C), and the total training sessions ($\beta = -0.035$, $p < 0.001$, Figure 9D). The RST load showed a marginal effect as a potential moderator ($\beta = -0.002$, $p = 0.053$, Figure 10A). There was no significant linear relationship between the other 14 moderator variables and RST training effects (ESM Figure S4). Using simple linear regression to calculate all moderators revealed that statistically significant results were more frequently obtained with this approach (83.3%, 15/18) compared to the *metafor* package (22.2%, 4/18).

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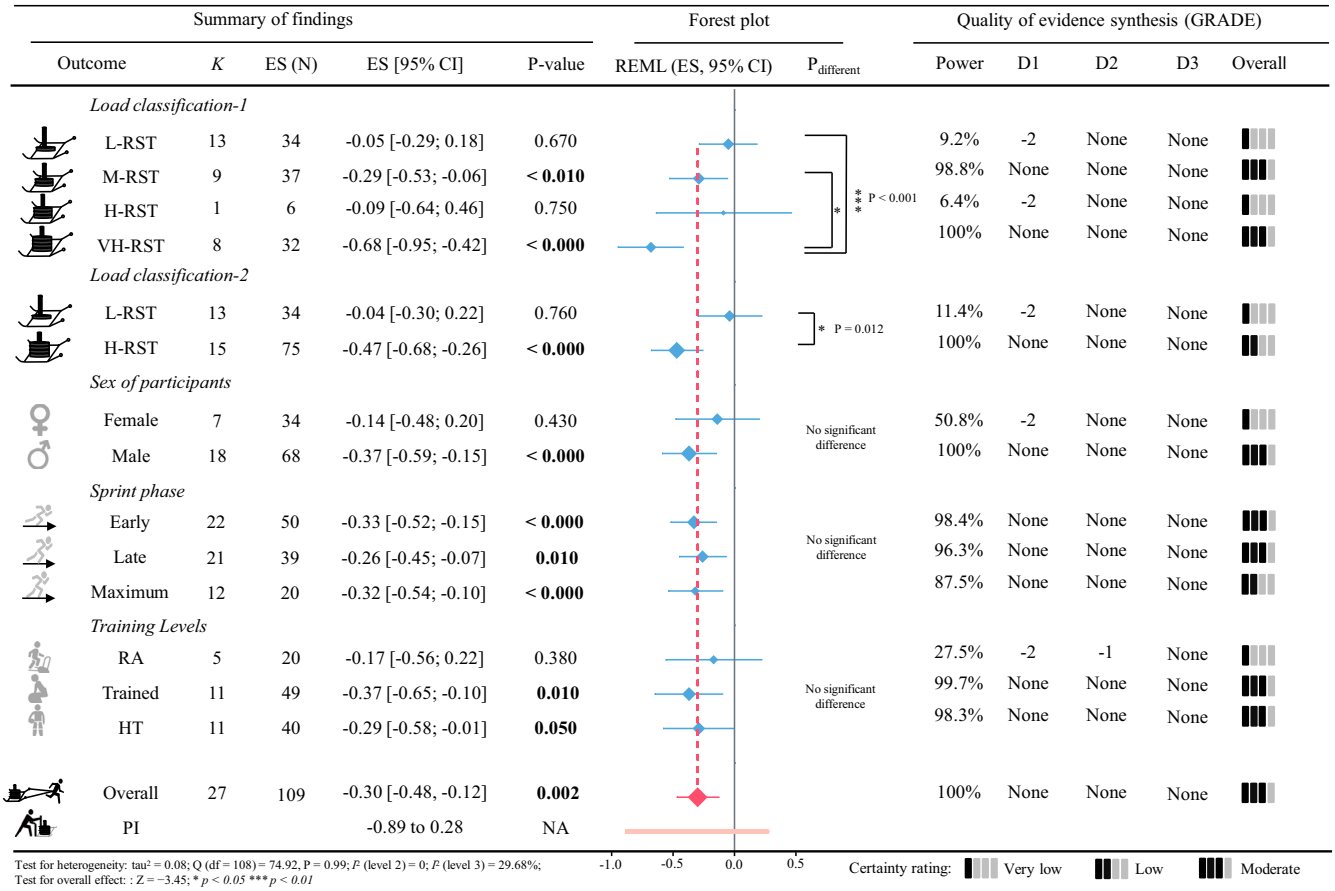

FIGURE 6 | Effect of RST versus UST on sprint performance subgroup analysis results, D1, whether the statistical power reaches 80% and the effect size has a clear direction; D2, whether $I^2 > 50\%$; D3, whether there is a risk of publication bias; L, light; M, moderate; H, heavy; VH, very heavy; RST resisted sprint training; UST, unresisted sprint training; RA, recreationally active; TR, trained, HT, highly trained, PI, prediction intervals; REML, restricted maximum likelihood; ES, effect size; ES (N), number of effect sizes; CI, confidence interval.

There were nonlinear relationships between participant age (Figure 9A), training session volume (Figure 10C), and total training volume (Figure 9E), exhibiting L-shaped, zigzag-shaped, and inverted U-shaped relationships, respectively. For the very heavy RST, there were inverted U-shaped relationships between the number of sessions and total training volume and the training effect (ESM Figure S3C). For moderate RST, an inverted U-shaped relationship was observed between total training volume and training effect (Figure 9G).

Meta-regression analysis on sprint mechanics showed significant increases in both maximum theoretical power ($\beta = 0.006$, $p = 0.005$, Figure 11A) and maximum power ($\beta = 0.004$, $p = 0.048$, Figure 11B) with higher RST loads.

3.6.2 | Two-Arm Comparison

Compared to UST, among the 18 continuous moderators of sprint performance, RST effects on sprint performance were significantly moderated by the total number of training sessions ($\beta = -0.045$, $p = 0.003$, Figure 12D) and RST load ($\beta = -0.007$, $p = 0.004$, Figure 10D). Significant linear meta-regression results were also obtained using the upper ($\beta = -0.007$, $p = 0.005$) and lower limits ($\beta = -0.006$, $p = 0.004$) of each RST loading range. There was no significant linear relationship between the other 14 moderator variables and the RST training effect (ESM Figure S5).

Only total training volume showed an inverted U-shaped relationship with the training effect (Figure 12E). There was no significant linear or nonlinear relationship between other variables and the training effect (Figure 12A–D,F–H, ESM Figure S6). Notably, light loads did not show any significant difference compared to UST at any training volume.

In meta-regression analyses using sprint mechanics as the outcome, maximum theoretical force ($\beta = 0.010$, $p = 0.023$, Figure 11D) increased significantly with higher RST loads. P_{max} emerged as a potential moderator variable with marginal effects ($\beta = 0.007$, $p = 0.096$, Figure 11E).

4 | Discussion

The choice of RST load magnitude has been a topic of ongoing debate in sport science, with the optimal training parameters still unresolved. This study aimed to address these gaps through a rigorous systematic review and meta-analysis, emphasizing methodological quality and attempting to refine the approach

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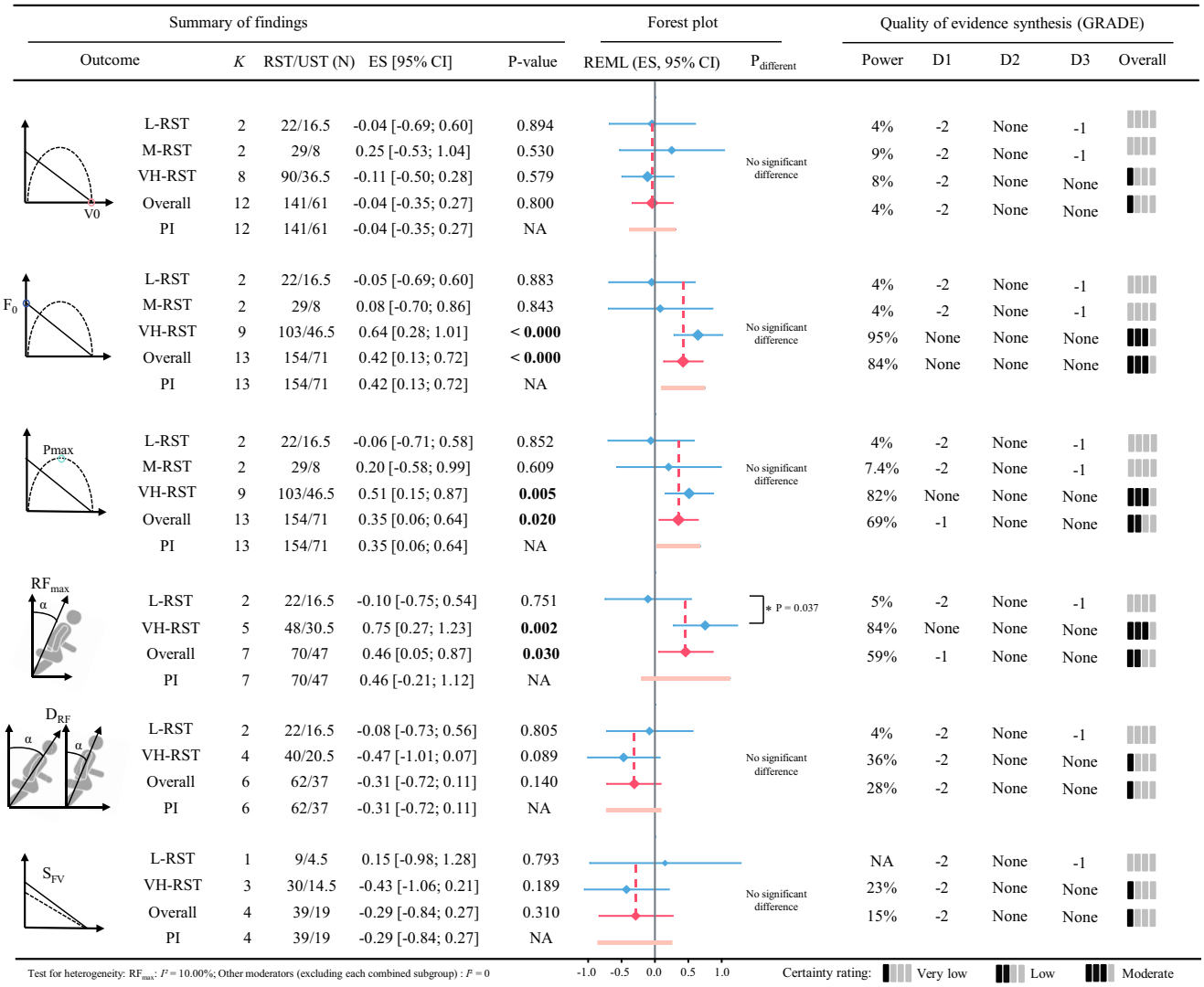

FIGURE 7 | Effect of RST versus UST on sprint mechanics subgroup analysis results, D1, whether the statistical power reaches 80% and the effect size has a clear direction; D2, whether $I^2 > 50\%$; D3, whether there is a risk of publication bias; L, light; M, moderate; H, heavy; VH, very heavy; RST, resisted sprint training; RA, recreationally active; TR, trained; HT, highly trained; PI, prediction intervals; REML, restricted maximum likelihood; ES, effect size; CI, confidence interval; RST/UST (N), sample size for RST and UST; v_0, maximal theoretical running velocity; P_{max}, maximal power; F_0, maximal theoretic al horizontal force; RF_{max}, maximal ratio of force; DRF, decrease in the ratio of force; S_{FV}, slope of the linear force-velocity relationship.

to defining and quantifying RST. To this end, we employed a three-level meta-analysis and a Bayesian arm-based multilevel network meta-analysis to assess the effects of different RST loads on sprint performance and mechanical characteristics. Additionally, we examined how various training parameters and participant characteristics impacted sprint performance. Importantly, the single-arm analyses are presented solely as exploratory and supportive evidence, whereas the main conclusions are based on the two-arm and network meta-analyses.

The main results were as follows:

1. Very heavy loads of RST (≥80% BM or ≥50% v_{dec}) were the most effective for improving sprint performance, outperforming light loads (<20% BM or ≤10% V_{dec}) in both two-arm (RST vs. UST; Figure 6) and multi-arm comparisons (intercomparison of different loads of RST, UST, and control group; Figure 8, Table 2).

2. With sufficient statistical power (100%), RST was more effective than UST for improving sprint performance (Figure 6), and this effect of RST increased significantly with higher %BM loads. Meta-regression results revealed that 20.7% BM load is the minimal load necessary to achieve superior adaptations with RST compared to UST (Figure 10D).

3. In the single-arm comparison, RST significantly improved sprinting mechanical characteristics including F_0, P_{max}, RF_{max}, DRF, and S_{FV}, with very heavy loads showing the greatest improvements (Figure 4). RST load was a significant moderator for F_0 and P_{max}. Compared to UST, only very heavy loads significantly improved F_0, P_{max}, and RF_{max}, while light and moderate

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loads had similar effects to UST (Figure 7). Additionally, F_0 improvements were greater with higher RST loads and more effective than UST when the load exceeded 54.3% BM (Figure 11).

4. RST significantly improved sprint performance from the acceleration to the maximum velocity phase (Figures 3, 5, and 6). However, when excluding early acceleration phases (e.g., 5–10, 10–20 m), RST had only a small effect on sprint performance and showed no more advantage over UST (Figure 5).

5. Light RST had a greater effect on sprint performance in recreationally active groups but was less effective for trained and highly trained individuals (Table 1). In contrast, moderate and very heavy RST were more effective for trained and highly trained groups but had a smaller effect on recreationally active groups (Table 1).

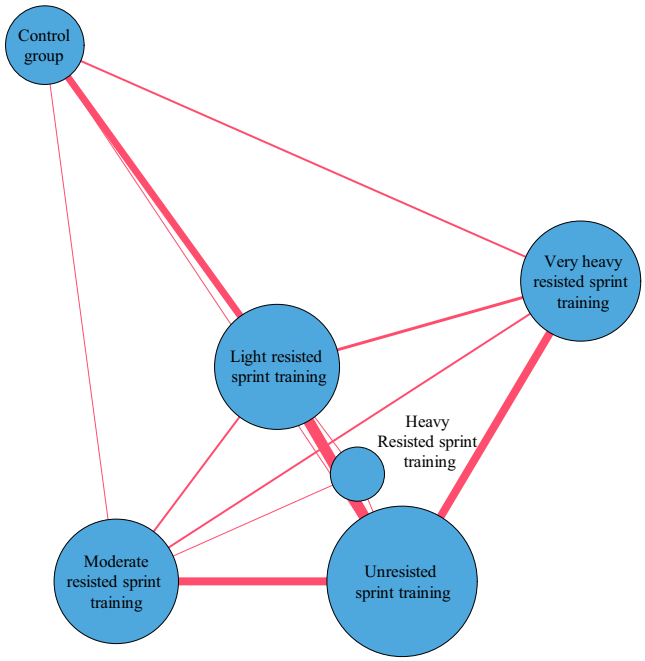

FIGURE 8 | Network diagram illustrating the pairwise effect sizes calculated across four RST load categories (light, moderate, heavy, very heavy), unresisted sprint training, and the control group. The analysis included direct and indirect pairwise effect sizes derived from 39 studies. The relative size of the nodes and the relative thickness of the connecting lines indicate the frequency of measurements within a given intervention and the number of direct comparisons between interventions, respectively.

6. The effect of RST was consistently greater in males than in females (Figures 3 and 6).

7. In the single-arm comparison, RST significantly improved sprint performance in participants aged 12 and older (Figure 9A). In the two-arm comparison, age did not moderate the differences between RST and UST (Figure 12A).

8. In the single-arm comparison, longer training duration, higher frequency, and more total training sessions were associated with greater sprint performance improvements through RST (Figure 9B–D). However, compared to UST, the effect of RST on sprint performance significantly increased only with the total number of training sessions (Figure 12D).

9. There was an inverted U-shaped relationship between total training volume and training effect (Figures 9E and 12E). As heavy RST studies typically used shorter distances than lower-load RST studies, large loads with low volumes may yield similar benefits to lower loads with higher volumes.

4.1 | RST Pre Versus Post Comparisons

Consistent with previous meta-analyses [8–10, 18, 41], our findings from single-arm analyses confirm the effectiveness of RST for improving sprint performance. The prediction interval for pooled effects did not overlap with the line of no effect (Low GRADE), strengthening confidence that future chronic RST interventions (≥4 weeks) are also likely to produce meaningful sprint performance improvements. However, because single-arm comparisons may introduce bias in meta-analyses [118, 119], and given that many RST studies [31, 120–124] do not include either control or UST groups, these studies should be interpreted cautiously. Nevertheless, their inclusion remains consistent with previous RST meta-analyses [8–10, 18, 41], and provides supportive insight alongside the more rigorous comparative analyses.

TABLE 2 | Summary of network meta-analysis results from 36 studies using the unresisted sprint training as a reference.

Comparison of RST for different loads	Effect size [95% CrI]		SUCRA (%)	P-score
UST	—	0.17 [−0.05, 0.39]	22%	0.22
CG	0.13 [−0.21, 0.48]	0.30 [−0.01, 0.62]	9%	0.09
L-RST	−0.17 [−0.39, 0.05]	—	56%	0.60
M-RST	−0.20 [−0.47, 0.05]	−0.03 [−0.33, 0.26]	61%	0.57
H-RST	−0.18 [−0.68, 0.31]	−0.02 [−0.52, 0.49]	55%	0.54
VH-RST	−0.51 [−0.77, −0.26]	−0.34 [−0.63, −0.06]	97%	0.98

Abbreviations: CG, control group; CrI, credible intervals; H, heavy; L, light; M, moderate; P-score, Ranking the effect on sprint performance; RST, resisted sprint training; SUCRA, Ranking the effect on sprint performance; VH, very-heavy.

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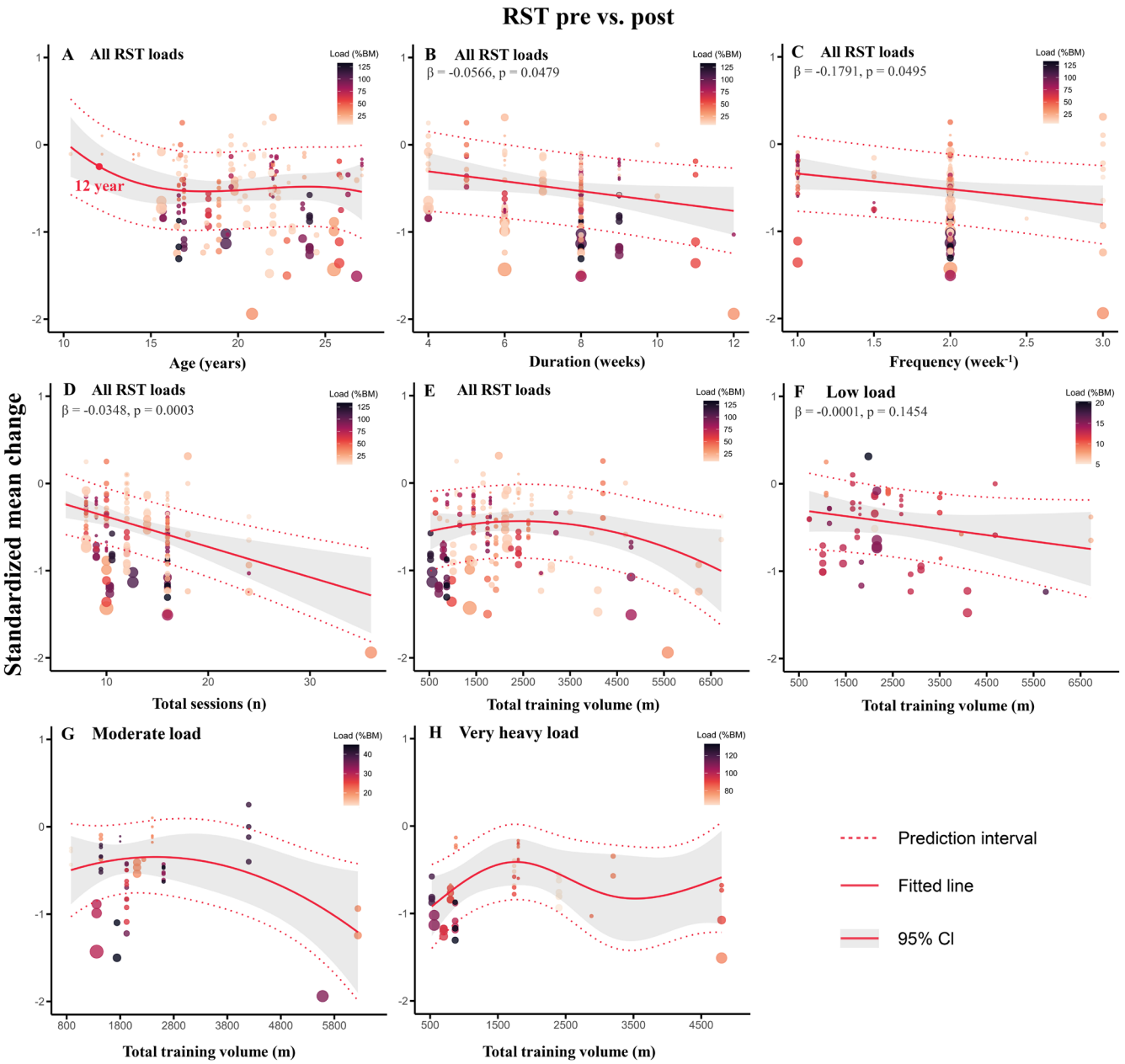

FIGURE 9 | Pre- versus post- RST analysis showing the effects of different moderators on the intervention outcomes. Panels A, B, C, D, and E depict the impact of participant age, duration, frequency, total sessions and total training volume on the effectiveness of the RST intervention across all loads. The red dot in panel A indicates that RST significantly improved sprint performance when participants were older than 12 years. Panels F, G, and H illustrate the effect of total training volume on RST outcomes under light, moderate, and very heavy load conditions. The gradient of the data points represents the RST load magnitude, with darker colors indicating heavier loads. RST, resisted sprint training, CI, confidence interval, BM, body mass.

Subgroup analysis based on load classification 1 (Figure 1) showed significant sprint performance improvements across all four load classifications (low, moderate, heavy, and very heavy). However, although the point estimate for very heavy loads was the largest, no statistically significant differences were detected between categories. Similarly, load classification 2 (Figure 1) revealed comparable improvements between light and heavy loads, with no clear relationship between increased sprint performance and higher RST loads ($p = 0.053$). These findings from this single-arm analysis suggest that RST at any load may enhance sprint performance. However, 71.4% of studies included additional forms of training (e.g., strength training, small-field games) during the intervention period. As such, the presence of confounding training components highlights the need for caution when attributing improvements solely to RST load magnitude. Without appropriate comparator groups, it remains unclear whether the improvements were due to RST alone; therefore, we provided a more nuanced analysis of load-specific effects in the two-arm comparison.

Pooled results showed that RST significantly improved various key mechanical variables underpinning sprint performance, including F_0, v_0, P_{max}, RF_{max}, S_{FV}, and DRF. However, the significant effect of RST on v_0 should be interpreted with

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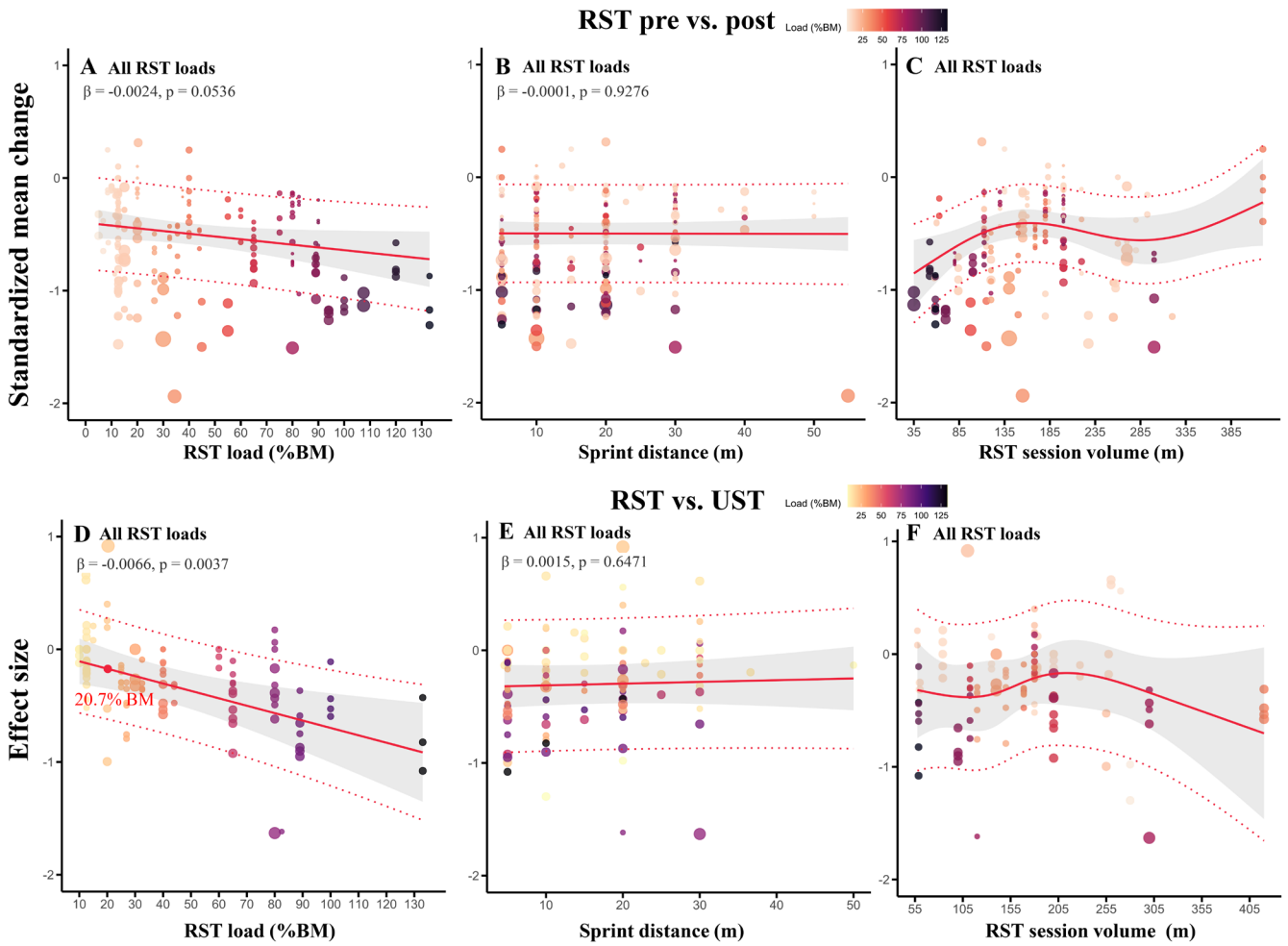

FIGURE 10 | The effect of RST load amplitudes, sprint phases, and session volume on RST intervention outcomes. Panels A, B, and C depict a pre–post comparison of RST. Panels E, F, and G depict a comparison of RST to UST. The gradient of the data points represents the RST load magnitude, with darker colors indicating heavier loads. RST, resisted sprint training; UST, unresisted sprint training; BM, body mass.

caution due to low statistical power (18%), and subgroup analyses based on load were not statistically significant. Further subgroup analyses using load classification 1 (Figure 1), which categorizes loads into four groups, revealed that improvements in sprint mechanical characteristics were primarily driven by very heavy loads. Specifically, F_0 increased significantly as RST loads increased. Considering that very heavy loads produced the greatest improvements in sprint performance, it seems likely that RST mainly improves sprint performance by increasing the force component of the force-velocity profile and improving power output, with a more horizontal force application [1, 14]. However, no significant differences emerged between the effects of different loads on sprint mechanical characteristics. This may be attributed to the limited number of studies examining light ($n = 3$) or moderate loads ($n = 2$), with most research focusing on very heavy loads ($n = 14$). The overall statistical power of these studies was low (5%–44%), and the level of evidence was rated as low (unranked to low GRADE). Further research should explore the specific effects of different RST loads on sprinting mechanical characteristics.

Consistent with previous findings [8, 9, 40, 41], our results indicated that the effect of RST overall enhancing sprint performance mainly originated from changes in the early acceleration phase (0–5 m). This could be due to its ability to emphasize the development of associated phase-specific adaptations. Examples are promoting a more forward-leaning posture during acceleration [14, 54], facilitating increased horizontal force application [29, 30] and resultant ground reaction forces [30, 54]. These effects are further supported by observed improvements in sprint mechanical characteristics (Figure 6). However, unlike previous studies [8, 9, 18, 41], this meta-analysis revealed significant improvements in sprint performance at 5–10, 10–20, and 20–50 m. While a recent study [9] reported significant RST effects at 10–20 m but not beyond 20 m, the significant effects observed here may stem from including more original studies. Nonetheless, the statistical power for RST effects across all sprint segments (sprint phase classification 2; Figure 1) remained below 80%, and the overall evidence level was generally low. Notably, the low power highlights the potential for missed effects, particularly for smaller or less-studied outcomes, and underscores the need for cautious interpretation and further research.

Subgroup analyses based on sex showed that RST led to significantly greater improvements in males than females. This aligns

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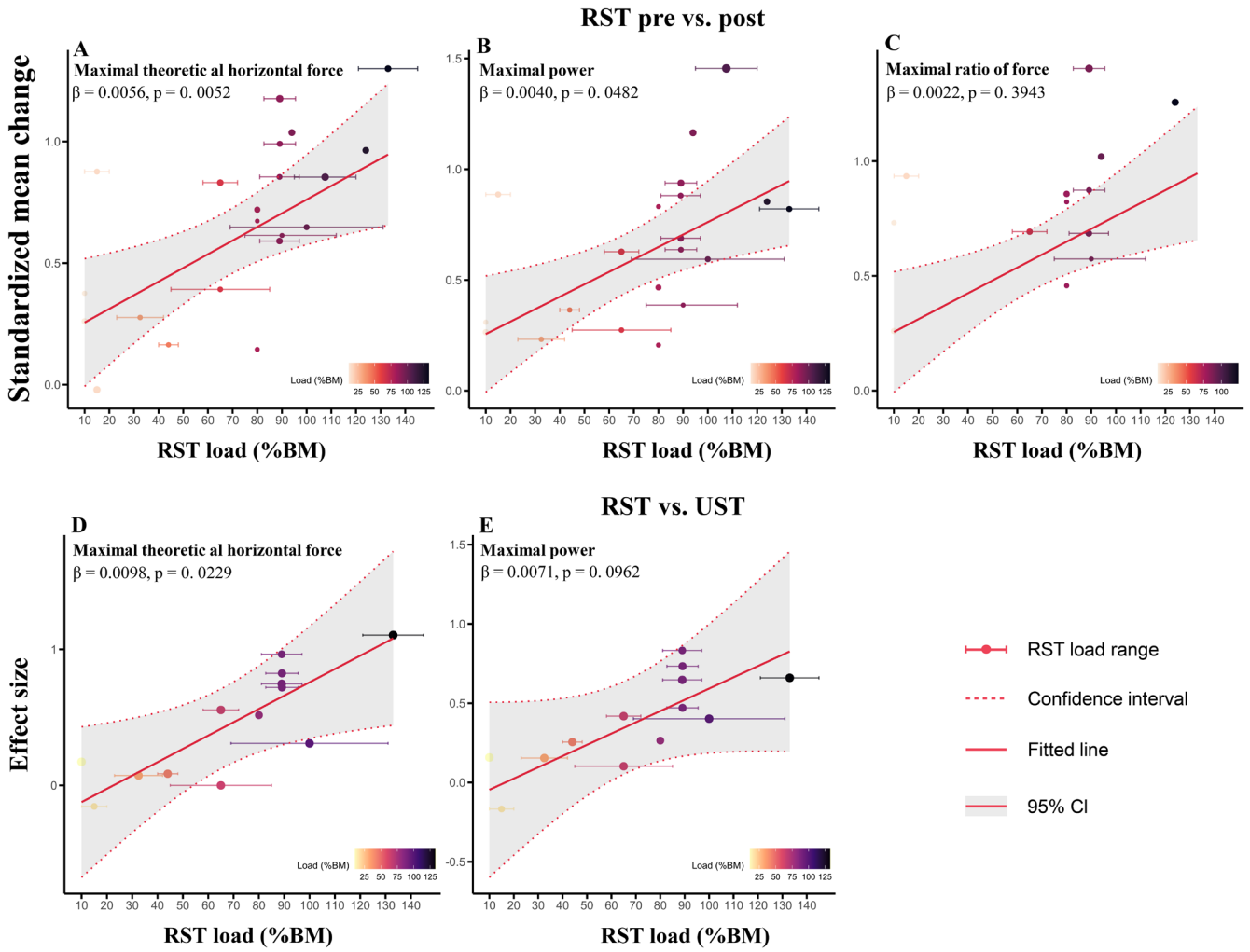

FIGURE 11 | The figure illustrates a linear meta-regression comparing sprint mechanics between single-arm and two-arm conditions. Panels A and D illustrate the relationship between RST load magnitude and maximum theoretical strength. Panels B and E illustrate the relationship between RST load magnitude and maximum power. Panel C illustrates the relationship between the RST load magnitude and the maximum ratio of force. The gradient of the data points represents the RST load magnitude, with darker colors indicating heavier loads. RST, resisted sprint training; UST, unresisted sprint training; CI, confidence interval; BM, body mass.

with findings from Alcaraz et al. [41] for the early acceleration phase. Several recent studies [125, 126] have suggested that females may benefit from shorter sprint distances and lower RST loads (~10% less BM) to minimize neuromuscular fatigue while maintaining comparable step counts and joint kinematics to males [125]. A practical interpretation is that RST loads in mixed-sex groups should be adjusted, with females using lighter loads to optimize training outcomes. However, this recommendation should be interpreted with caution due to the limited number of studies including female participants. Apart from sex differences, training experience may also influence RST adaptations [6, 9, 10, 39, 41, 45]. Current evidence suggests that light loads are most effective for recreationally active individuals, while moderate and very heavy loads benefit trained athletes. Therefore, beginners should start with light loads, progressively increasing resistance after familiarization. For advanced athletes, heavier loads may provide a more effective training stimulus, depending on the training objective. Similarly, RST load selection should consider participants' characteristics, potentially via their individual load-velocity profiles [127]. Future research should explore how sex and training level interact to influence RST adaptations.

Meta-regression analysis revealed that chronic RST significantly improves sprint performance in participants aged 12 and older. However, pre-peak height velocity (PHV) participants did not benefit from light to moderate RST loads (2.5%–10% BM, 10%–30% BM). In contrast, heavy or very heavy RST loads are suitable for trained post-PHV athletes. Both Cahill et al. [67, 68] and Derakhti et al. [128] found that using 25%–75% v_{dec} loads significantly improved sprint performance in athletes aged 15.7 (±0.5) to 16.9 (±0.8) years. Therefore, for pre-PHV athletes, it appears as though RST should primarily be utilized as an educational and familiarization tool rather than a recommended training method. It Implementing progressive loading and emphasizing proper sprint mechanics may facilitate their readiness for future RST application. For post-PHV athletes, load selection should depend on their training level.

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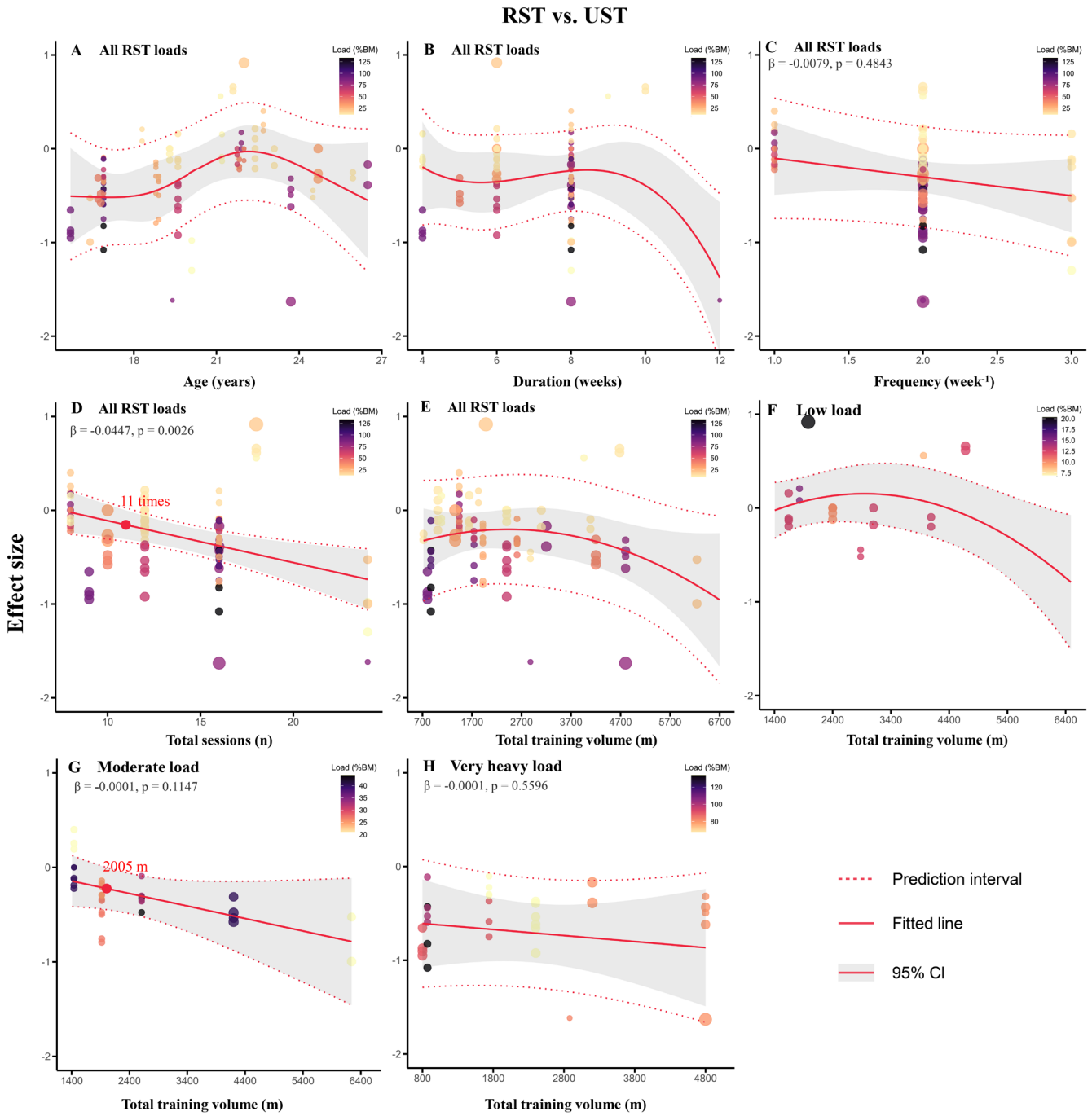

FIGURE 12 | RST compared to UST analysis showing the effect of different moderators on the intervention outcomes. Panels A, B, C, D, and E depict the impact of participant age, duration, frequency, total sessions and total volume on the effectiveness of the RST intervention across all loads. The red dot in panel A indicates that RST significantly improved sprint performance when participants were younger than 20 years. The red dot in panel D indicates that RST significantly improved sprint performance when total sessions were larger than 11 times. Panels F, G, and H illustrate the effect of total training volume on RST intervention effects under light, moderate, and very heavy load conditions. The red dot in panel G indicated that RST significantly improved sprint performance when the total training volume of moderate load was greater than 2005 m. The gradient of the data points represents the RST load magnitude, with darker colors indicating heavier loads. RST, resisted sprint training; UST, unresisted sprint training; CI, confidence interval; BM, body mass.

Another key finding was that longer RST duration, higher training frequency, and more total training sessions led to greater sprint performance improvements. However, even a short duration (4 weeks), once a week, and 6 sessions total intervention effectively enhanced sprint performance, suggesting a minimum effective RST volume. Meta-regression analysis showed that very heavy loads required fewer sessions and lower volumes to achieve maximal effects. For instance, Derakhti et al. [128] reported an effect size of 0.89 in 0–30 m sprint time using a 50% v_{dec} load over 4 weeks (9 sessions, totaling a RST volume of 800 m). Similarly, Lahti et al. [33] achieved an effect size of 1.23 for 0–20 m sprint time in trained rugby players with a 120% BM load over 30–75 m

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sessions (totaling 525 m). This contrasts with Alcaraz et al.'s [41] who suggested that a total RST volume > 2680 m would improve sprint performance better. Furthermore, meta-regression indicated a positive but non-significant relationship between low loads and total training volume. In contrast, moderate and very heavy loads exhibited an inverted U-shaped relationship, suggesting that sprint performance gains from high load, low training volume may be comparable to those achieved with low-load, high-volume training. This pattern differentiates moderate and very heavy load RST from light-load RST. However, it is worth noting that volume in sprinting research is typically reported as distance, while the actual training stimulus—time under load—varies with load. Heavier loads result in longer sprint durations for a given unit of distance, while lighter loads allow for shorter sprint times. Therefore, properly controlling for time under load may provide a clearer understanding of how loading influences different performance adaptations.

In summary, light, moderate, heavy, and very heavy RST loads all resulted in varying improvements in sprint performance and mechanical characteristics, highlighting load as an important moderator of RST-induced adaptations. This finding challenges the validity of previous meta-analyses that grouped all loads under a generic "RST" descriptor. Very heavy loads emerged as the most effective overall. However, for recreationally active individuals with limited sprint training experience, light loads are recommended as an initial step before implementing heavier RST, with very heavy loads reserved only for educational and familiarization purposes with this population.

4.2 | RST Versus UST Comparisons

In contrast to previous findings [8–10, 39, 41], our pooled meta-analysis indicated a greater enhancement in sprint performance following RST compared to UST (Moderate GRADE). The discrepancy between studies could be attributed to two main factors: (i) previous meta-analyses of RST used a conservative within-group (RST pre vs. post) correlation of 0.5, based on Cochrane handbook recommendations [85] or other formulas [39]. However, Grazioli et al. [62] reported sprint time pre-post correlations between 0.56 and 0.79. Based on this evidence, we opted to adopt a correlation of 0.6 and found our pooled and subgroup results robust during sensitivity analyses; and (ii) the use of a three-level model for pooling [87, 91, 93], which increased statistical power (100%) and retained more information from the original datasets, resulting in greater accuracy. Nonetheless, the prediction interval around the mean improvement (−0.89 to 0.28) suggests residual uncertainty, underscoring the need for cautious interpretation of our findings [84].

Unlike the single-arm comparison, the two-arm results unveiled performance differences between UST and RST. Very heavy RST significantly outperformed both light and moderate loads in improving sprint performance. Similarly, in classification 2 (Figure 1), heavy loads were more effective than light loads. Across all eight studies on very heavy loads, RST significantly outperformed UST, while the difference between RST and UST was minimal in the 13 studies examining light loads. Meta-regression further confirmed that there was a positive relationship between RST load and improvements in sprint performance. Specifically, loads exceeding 20.7% BM were significantly more effective than UST. As such, despite recent criticism of very heavy load RST [45], our evidence supports its effectiveness, with very heavy loads (≥ 80% BM or ≥ 50% v_{dec}) outperforming UST and lower loads.

The pooled results showed that RST significantly improved F_0, P_{max}, and RF_{max} compared to UST, with only very heavy loads achieving significant improvements in these parameters. These results appear to align with recommendations [50] that using an RST load that corresponds to roughly the middle of the horizontal force- and load-velocity spectrum (i.e., 50% v_{dec} [69%–96% BM in the referenced work]) may be an effective stimulus for broad sprint development in certain contexts. This contrasts with other suggestions of "optimal" loading, which are fairly constrained to quite low loading protocols, either based on instantaneous measures (based on potentially inaccurate quantification of friction and resistance [129]) and a combination of arguably misinterpreted acute evidence [41, 45]. These findings, alongside the significant positive correlation between F_0 and RST loading, suggest that very heavy RST is an effective stimulus for enhancing both sprint mechanical characteristics and sprint performance. RST significantly improved sprint performance across all phases, from early acceleration to maximum velocity, with the greatest effects observed with very heavy load RST. Light loads had no significant effect in any phase, while moderate loads were effective only in the early acceleration phase. However, segmented sprint data (5–10, 10–20, and 10–54.9 m) indicated that RST was not more effective than UST, suggesting again that adaptations may be phase-specific, and a primary benefit of RST (notably at high loads) is likely improvements in the early acceleration phase. Additionally, one important reason for not discussing non-significant variables is the limited number of studies (or effect sizes) available for analysis. This limitation prevents drawing reliable conclusions and underscores the need for further research to explore these variables in more detail.

The current data suggested no significant relationship between sprint distance and performance ($p = 0.56$). However, Hamad et al. [39] found a significant decrease in sprint performance enhancement with increasing sprint distance after excluding outliers in a linear meta-regression ($p = 0.04$). Their analysis relied on simple linear regression fit, which, as Borenstein et al. [95] pointed out, could induce incorrect p-values by failing to account for study heterogeneity and treating all effect values equally. More importantly, the meta-regression used weighted least squares to incorporate the weight of each effect size [88]. Our comparison of methods showed that simple linear regression yielded a significant p-value more frequently (83.3%) compared to 38.9% when using the *metafor* package. Importantly, Xu et al. [56] found that Hamad et al. [39] likely omitted several ESs in their linear meta-regression. When the correct statistical methods were applied, sprint distance did not significantly modulate sprint performance, regardless of outlier inclusion.

The results for participants' sex and training level aligned with the single-arm results. Compared to UST, all RST loads had a greater effect on males than females. While light load RST showed potential for recreationally active individuals, it did not significantly outperform UST. In contrast, moderate and very

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heavy loads were significantly effective for trained and highly trained groups, with the very heavy loads on the highly trained group reaching the ES of 1.11. These results emphasize that tailoring RST to the participant's training levelis particularly effective for high-level athletes.

Meta-regression analysis identified the total number of training sessions and RST load as significant moderating factors. More than 11 training sessions were required for RST to outperform UST. However, conflicting findings emerged. For instance, Luteberget et al. [130] found UST to be more effective than light RST load (12.4% BM) after 18 training sessions, while Derakhti et al. [128] noted a significant RST effect after just nine training sessions using a 50% v_{dec} load. These discrepancies suggest that very heavy loads may produce better results with fewer training sessions than lighter loads. Non-linear meta-regression further supported this (Figure 12E), showing very heavy loads with a total training volume of 800–870 m were as effective as moderate loads requiring 6240 m. These findings support Cahill et al.'s [52] suggestion to reduce single sprint distances as RST load increases to maintain an effective training stimulus. Additionally, light load RST required volumes like UST for comparable results, whereas moderate and very heavy loads achieved effectiveness at thresholds of 2005 and 800 m, respectively (see Section 4.1 for discussion on time spent at a given targeted velocity).

No significant trends were found for participant age, duration, frequency, or session volume. Meta-regression analysis on single RST sprint distances also yielded inconclusive findings due to substantial variation in training protocols. For example, Spinks et al. [131] implemented a progressive approach, increasing sprint distances from 5 to 25 m (a five-fold increase), making universal recommendations challenging.

In summary, very heavy loads showed the greatest advantages for sprint performance and mechanical characteristics.

4.3 | Effect of RST on Underlying Sprint Mechanics

Some studies [10, 23, 41, 45] suggest selecting RST loads that closely replicate sprinting technical movements (e.g., patterns and kinematics). However, our data show that very heavy loads, previously thought to disrupt sprinting patterns, were the most effective in improving sprinting performance (SURCA: 97%). Notably, the only study assessing both sprint kinematics and heavy-load RST effects [34] showed clear improvements in sprint performance post-training with no change in sprint patterns among professional football players. In contrast, light loads were the third most effective training method (SURCA: 56%) in the present study. Additionally, our findings from single-arm, two-arm, and multi-arm comparisons consistently showed that very heavy loads had a significantly greater effect on sprint performance than light loads. Therefore, when the primary goal is improving sprint performance, concerns about potential side effects on sprinting technique should not outweigh the benefits of selecting heavier RST loads.

Several studies [6–8, 10, 19, 23, 40, 51, 52] that suggest RST loads should not exceed 10% v_{dec} (~12.5% BM) often base this recommendation on anecdotal evidence rather than scientific proof. Cross-sectional studies [27, 43, 44, 99, 132] have emphasized that heavier RST loads (especially very heavy ones) alter sprint kinematics (i.e., reduced step frequency, step length, take-off time, increased ground contact time, greater knee flexion, and restricted hip motion). However, these cross-sectional studies are observational and do not establish causation [133, 134], nor do they provide insights into long-term training-induced effects. As with many training stimuli (e.g., environmental stress), acute (in-session) performance changes do not always predict long-term outcomes. For example, the sprint mechanical alterations observed during repeated sprint training sessions under severe hypoxic conditions [46] are not evident after a period of altitude training when tested in normoxic environments [135]. In addition, a recent cross-sectional study [48] found minimal kinematic differences between very heavy and lighter RST loads at equivalent running velocities. This suggests that observed alterations in running mechanics may stem from slower running speeds rather than the load itself. Once running speed was controlled for, resistive loads showed minimal impact on running patterns. Furthermore, Lahti et al. [34] found no significant changes in unrestricted sprint kinematics after prolonged very heavy RST, even during early acceleration or top-speed phases. Contrary to concerns about very heavy loads negatively affecting sprint performance, this study revealed significantly greater gains with increasing RST loads. Some studies have shown that both non-specific and specific training can improve sprint performance in untrained populations [19, 20, 22], with more specific methods required to elicit further improvements as they progress [22]. Most participants in the studies reviewed here possessed significant sprint training experience, perhaps leaving little room for further gains through UST alone [136, 137]. As such, one hypothesis is that while light loads may help maintain intermuscular coordination, they fail to provide sufficient stimulus for trained populations, especially high-trained athletes [20]. In contrast, heavier loads (especially very heavy ones) provide greater overload while preserving sprint-specific coordination [1, 14, 138]. This effect likely results from increased recruitment of active muscles and/or reduced co-activation of antagonists [19, 20]. Note these interpretations are obviously constrained to the data available, and better controlling for aspects like training volume, and more detailed explorations of "phase specific" adaptations (e.g., lighter loading targeting late acceleration phase adaptations) would help further clarify the moderating effects of load on RST-induced adaptations.

Combining existing research with our pooled results, we conclude that heavier loads, especially very heavy loads, represent a unique training approach that bridges the gap between specific and non-specific training. Very heavy load RST not only directly improves sprint performance but can also enhance strength qualities that transfer effectively to sprinting [19, 20]. Nonetheless, loading application is context-specific. As Alcaraz et al. [41] emphasized, there is no one-size-fits-all "optimal load" for RST. The choice of load should align with the athlete's mechanical profile, training level, and objectives. For athletes with unstable sprinting technique and/or lower training experience, light load RST is a more suitable starting point, with heavier loads introduced progressively. Conversely, for highly trained athletes with stable sprinting technique, heavier and very heavy loads are more effective training options. Accordingly, while

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heavier loads appear to be clearly beneficial within this study, there is practical interest and utility in exploring a variety of loading parameters, depending on the individual and goal.

4.4 | Limitations

Of the 49 studies included in this systematic review, two [68, 139] employed a sled-push method. While sled pushing has a similar impact on sprint performance as sled pulling, it may disrupt proper sprinting mechanics due to restricted arm movements [52]. However, as discussed, RST primarily enhances the early acceleration phase, and fitting the mechanics of this phase is crucial for improving sprint performance [1, 14]. Intuitively, it seems unlikely that the brief duration of resisted sled-push training would negatively influence sprint technique, especially given the significant time athletes spend in unresisted running [51]. Furthermore, this study included extensive segmented data such as RST load and training volume, but only mean values were analyzed in the meta-regression. Although sensitivity analyses produced robust results, caution should be exercised when interpreting findings derived from segmented data. For example, while the model results remained unchanged, applying robust variance estimation in the meta-regression adjusted the effect load of RST exceeding UST from 20.7% BM to 19.7% BM. Another limitation is the low representation of female participants in the studies reviewed. Therefore, subgroups analyses labeled as "female" included some male participants, which may compromise the accuracy of sex-based subgroup comparisons. Moreover, although we categorized RST loads into four classifications, studies on heavy loads were limited, making the classification uneven. The lack of studies on moderate loads further restricts our ability to examine potential dose–response relationships, requiring cautious interpretation.

All studies included in this systematic review were classified as high risk of bias (Rob-2), primarily due to inadequate allocation concealment and insufficient reporting on randomization methods in all but two studies. Even when non-randomized assessment scales were used, 44.90% of studies still posed a serious risk. Additionally, 18 studies had sample size attrition. Hamad et al. [39] reviewed many of the same studies included here but reported markedly different results. They classified several non-randomized studies as low risk and failed to adequately address studies with sample size attrition. By following the appropriate evaluation method (Rob-2) [71], we found all studies to be at high risk of bias. While some aspects, such as the inability to obscure treatment conditions, are inherent to RST research, this underscores the need for future studies to improve methodological quality. Particular emphasis might be placed on robust randomization procedures and adherence to intention-to-treat analysis. Moreover, chronic (≥4 weeks) very heavy load RSTs may take 1–4 weeks post-training to fully manifest their effects on sprint performance [32]. Future studies should consider adding multiple follow-up time points after interventions to capture possible delayed gain effects and retention rates.

Considering the inconsistency observed between very heavy RST and UST in the network meta-analysis (i.e., the discrepancy between indirect and direct comparisons), we believe this is largely attributable to the fact that most evidence in the network comes from simple two-arm comparisons of RST vs. UST, with few studies including multiple RST loads, UST, and control groups simultaneously. Therefore, the probabilistic ranking of interventions based on the current network should be interpreted with caution. We also emphasize that, although this study employed multiple analytical approaches that produced largely consistent results, these findings should still be interpreted carefully. The relationship between RST load and performance remains complex, making it difficult to draw a universal conclusion across all training contexts.

In addition, the "highly trained athletes" included in this review were primarily professional team-sport athletes. Two studies [32, 35] have shown that very heavy RST can be effective for trained sprinters, but whether these findings extend to elite-level sprinters remains uncertain and warrants further investigation. Most interventions in this study relied on sled-based RST (43 studies, 87.6%). Conducting subgroup analyses based on training modality may introduce bias due to the small sample sizes for alternative training methods and result in insufficient statistical power. Although we believe that training modality is unlikely to be a primary determinant, and that load represents the more critical factor, future research is needed to clarify distinctions between modalities.

5 | Conclusion

Our results bring a more nuanced understanding of RST, which has often been viewed as a single training modality in previous meta-analyses, without considering key moderators such as load magnitude, athlete level, or training specifics. The results from two-arm and multi-arm comparisons indicate that very heavy load RST is more effective than light loads and UST for improving sprint performance. Meta-regression analysis indicated that RST tended to be more effective than UST when the RST load exceeded 20.7% of body mass; however, this finding should be interpreted with caution, as it may partly reflect the distribution of available data. Most of this improvement appears attributed to the early acceleration phase (especially 0–5 m), which is also evident in greater enhancements in key mechanical variables such as F_0, P_{max}, and RF_{max}, notably when using very heavy loads. Accordingly, when early acceleration was excluded (e.g., 5–10, 10–20 m), RST showed a more modest effect on sprint performance and did not outperform UST. This confirms the overall interest in high loads as a broad training stimulus and likely highlights the longitudinal transfer of loading based on its specificity to characteristics underlying specific phases.

6 | Perspective

To improve sprint performance, we recommend that trained athletes, especially those at advanced levels, incorporate very heavy RST (≥80% body mass or ≥50% v_{dec}) for a minimum of 4 weeks. Each resisted sprint should cover 7.5–30 m, with training sessions totaling 45–135 m per session and an overall training volume of 800–1440 m across the intervention period. For athletes with inconsistent sprinting techniques or lower training levels, light loads (≤10% v_{dec}, <20% BM) may be a more

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appropriate starting point, with a gradual progression to heavier loads. While very heavy loads (≥80% body mass or ≥50% v_{dec}) demonstrated significant effectiveness in this study, individualized responses should be carefully considered. Future studies should focus on tailoring RST loads and volumes to align with each athlete's specific training status and development level.

Author Contributions

Kai Xu performed all the analyses, visualized the data, and wrote the first draft of the manuscript. All authors edited and revised the manuscript and approved the final version of the manuscript.

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Ethics Statement

The authors have nothing to report.

Consent

The authors have nothing to report.

Conflicts of Interest

Kai Xu, Ivan Jukic, Matt R. Cross, Dylan Shaun Hicks, MingYue Yin, WenJing Tang, Yanfeng Li, Zhide Liang, Ran Wang, Jean-Benoît Morin, and Olivier Girard declare no conflicts of interest.

Data Availability Statement

The datasets generated and/or analyzed during the current review are available in tables, Supporting Information, and at the Open Science Framework (osf.io/fu82d). All other data are available upon request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Appendix S1:** sms70182-sup-0001-AppendixS1.docx. **Appendix S2:** sms70182-sup-0002-AppendixS2.R. **Data S1:** sms70182-sup-0003-DataS1.xlsx. **Data S2:** sms70182-sup-0004-DataS2.xlsx. **Data S3:** sms70182-sup-0005-DataS3.xlsx.

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